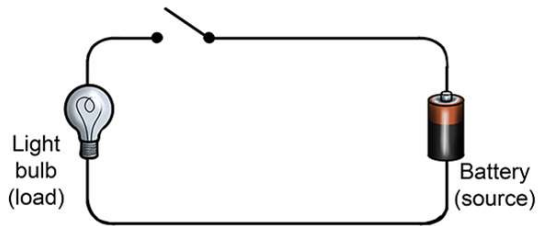
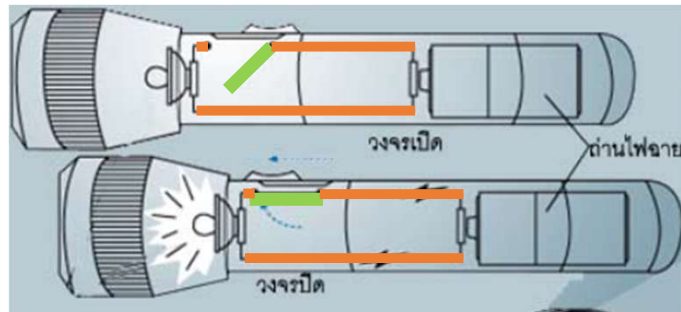
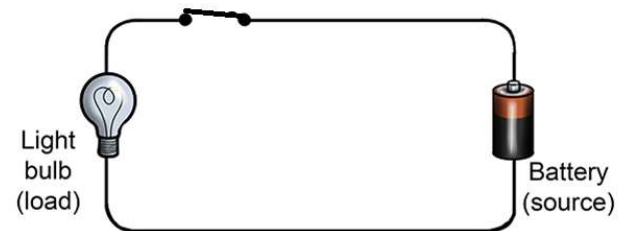
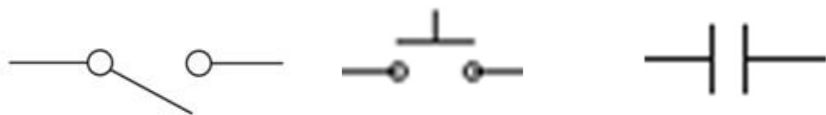


การควบคุมสวิตช์ Switch Control



วงจรเปิด (Open Circuit)

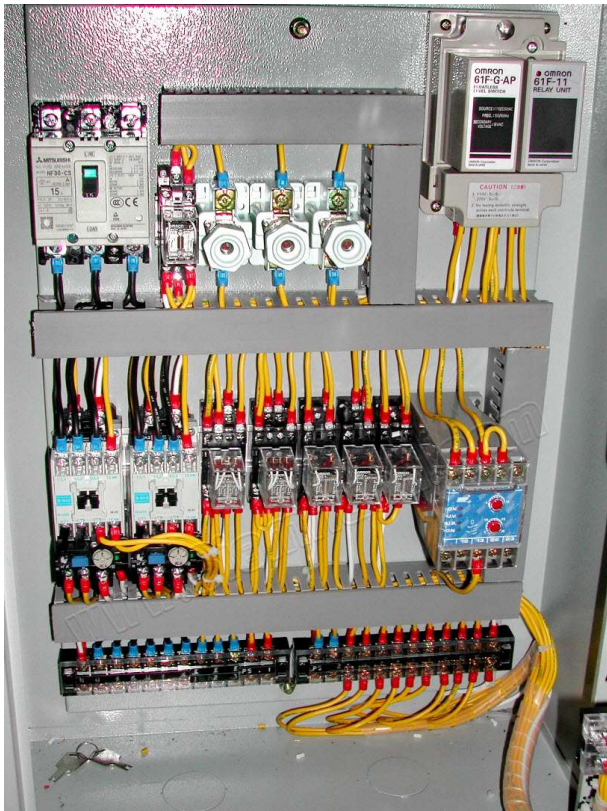
- กระแสไฟฟ้าไม่สามารถไหลผ่านได้ ทำให้อุปกรณ์ไม่ทำงาน
- หน้าสัมผัสไม่เชื่อมต่อกัน (Open Contact)



วงจรปิด (Close Circuit)

- หน้าสัมผัสเชื่อมต่อกัน (Close Contact)
- กระแสไฟฟ้าไหลในวงจรได้ ทำให้อุปกรณ์ทำงาน





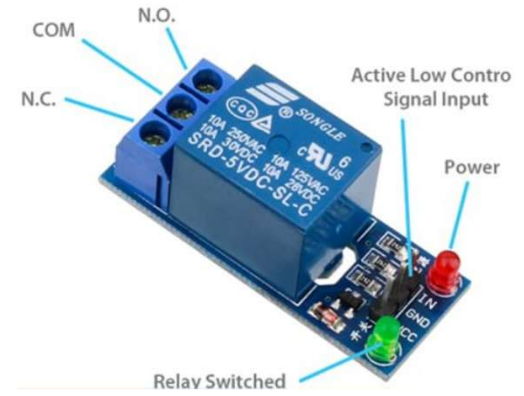
Normally Open Contact



Normally Closed Contact



Changeover Contact

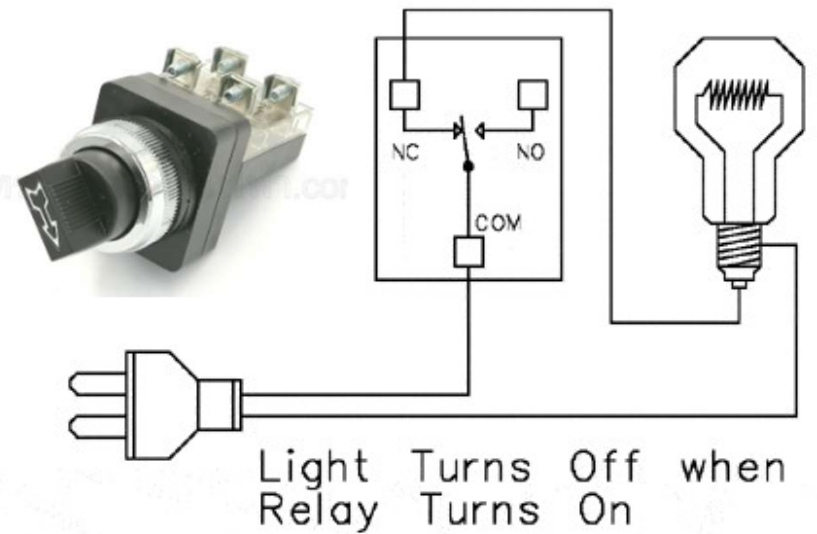
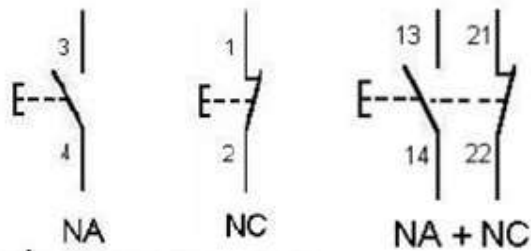
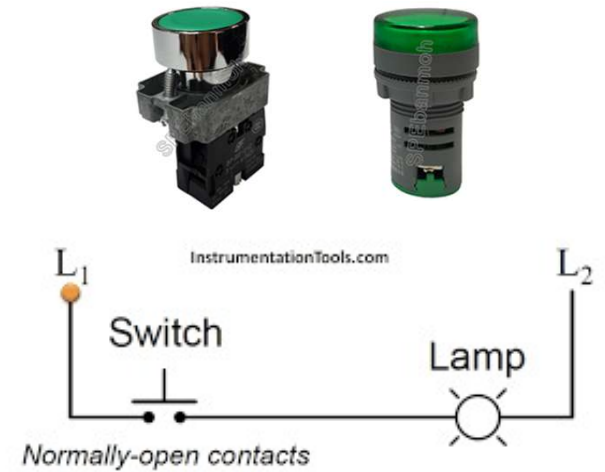
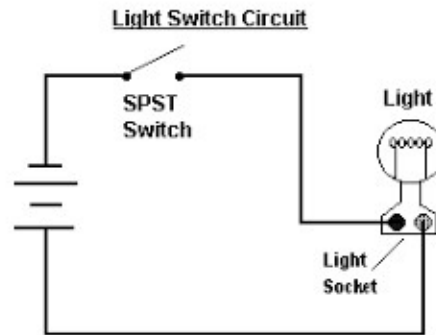




หมุนปุ่มมาทาง MAN ตั้งเปิดปิด จากสวิตช์หน้าตู้

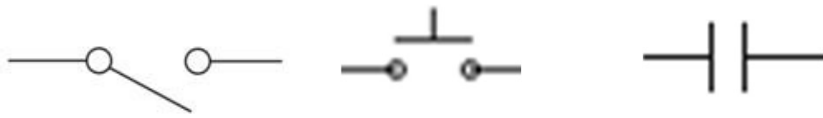


หมุนปุ่มมาทาง AUTO รอเวลาที่ตั้งไว้ จาก timer



วงจรเปิด (Open Circuit)

- กระแสไฟฟ้าไม่สามารถไหลผ่านได้ ทำให้อุปกรณ์ไม่ทำงาน
- หน้าสัมผัสไม่เชื่อมต่อกัน (Open Contact)



A Normal Open contact (NO)

ปกติหน้าสัมผัสเปิด

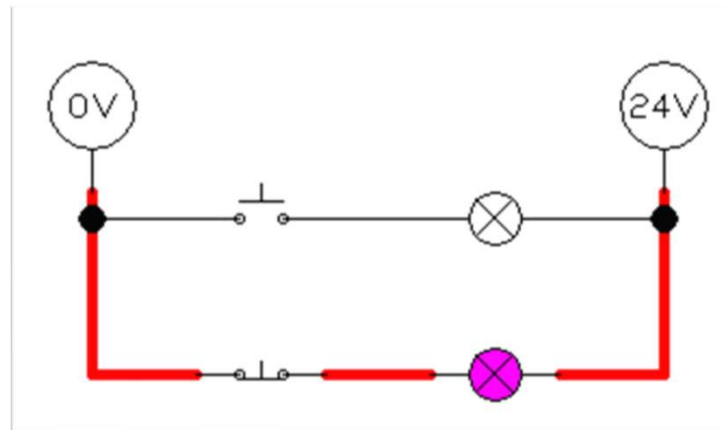
วงจรปิด (Close Circuit)

- หน้าสัมผัสเชื่อมต่อกัน (Close Contact)
- กระแสไฟฟ้าไหลในวงจรได้ ทำให้อุปกรณ์ทำงาน



A Normal Closed contact (NC)

ปกติหน้าสัมผัสปิด



Pressure se...

Analog pres...

Analog flow ...

Pneumatic t...

Displaceme...

Voltmeter

Ampere meter

Indicator light

Buzzer

Relay

Relay with ...

Relay with ...

Relay counter

Valve sole...

Proportional...

Starting curr...

Electrical co...

Electrical co...

Function ge...

Setpoint val...

Pushbutton ...

Detent swit...

Pushbutton ...

Detent swit...

Detent swit...

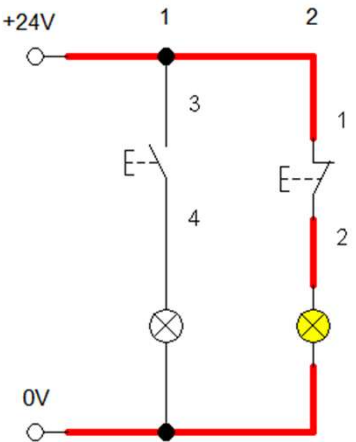
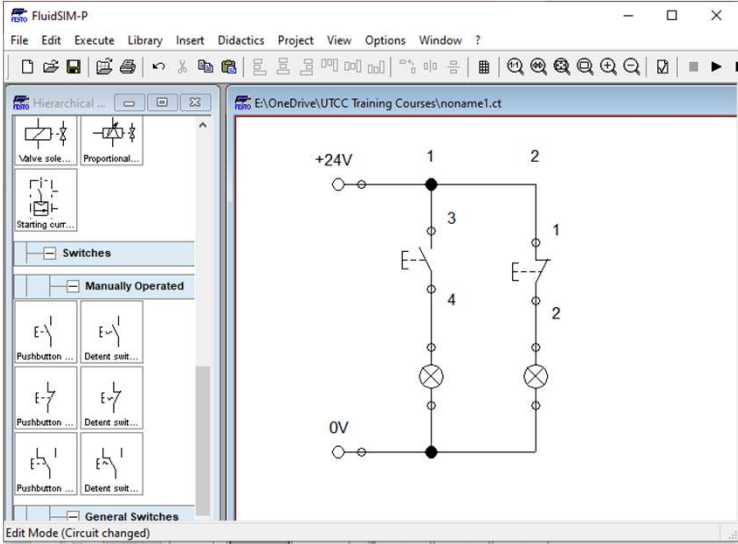
Pushbutton ...

Detent swit...

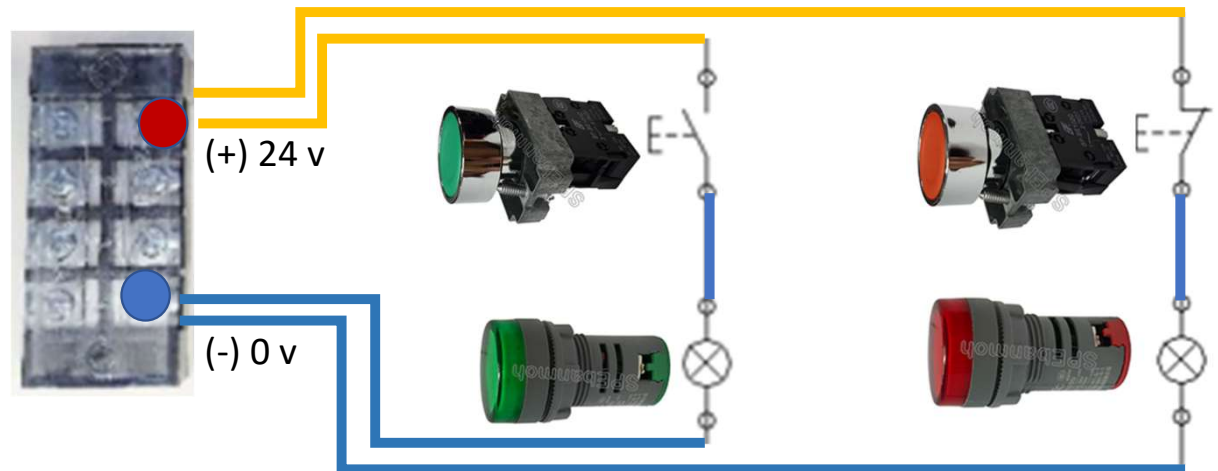
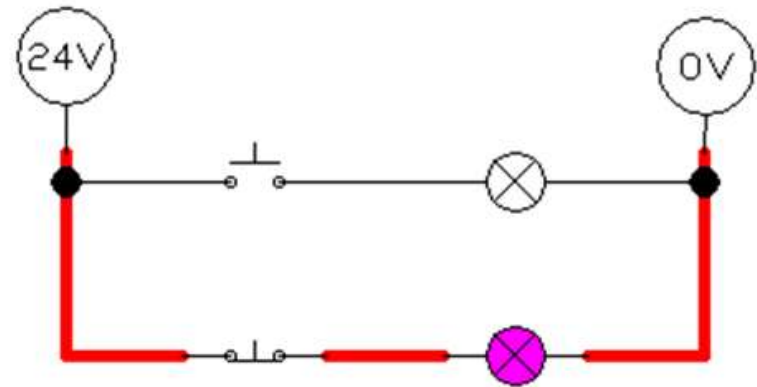
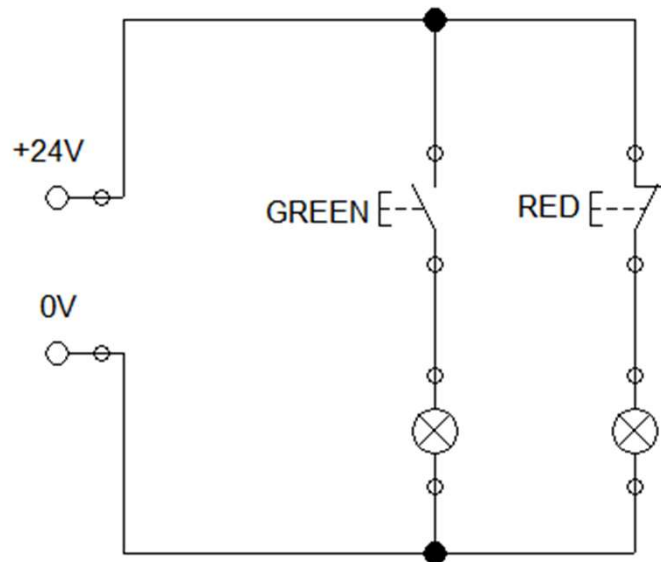
Make switch

Break switch

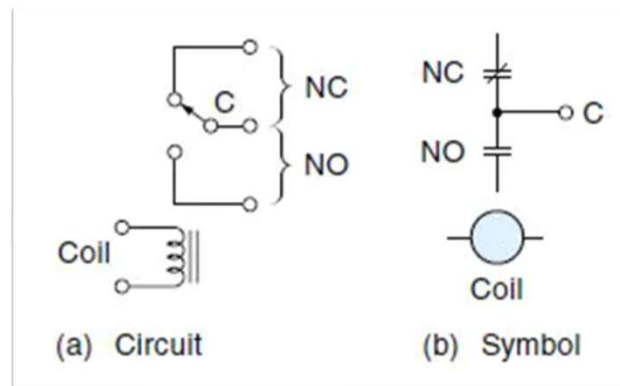
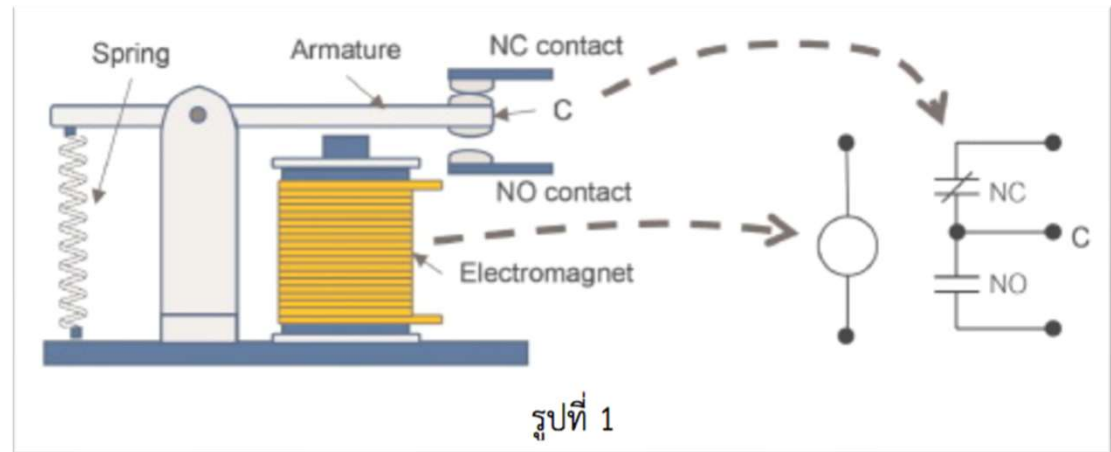
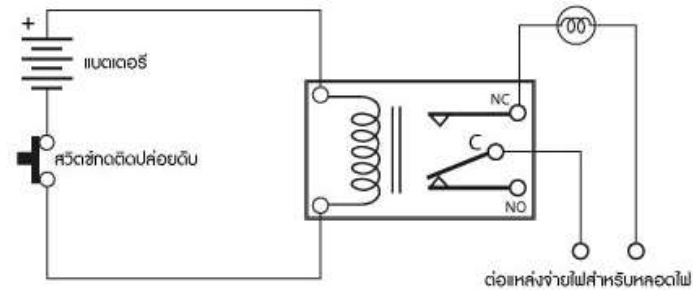
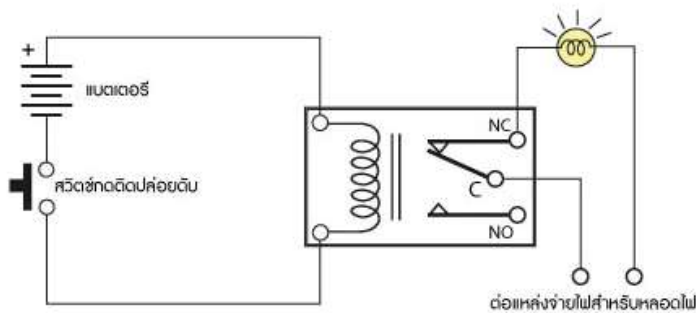
Changeover...

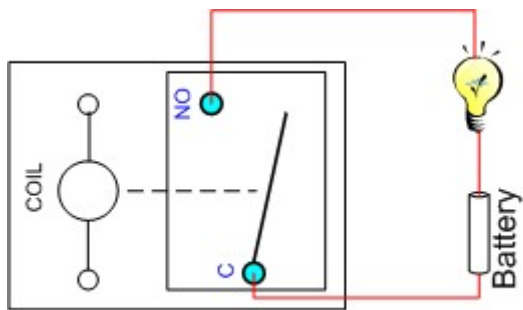


Basic 01 - Switch Control (NO-NC) [youtube](#)

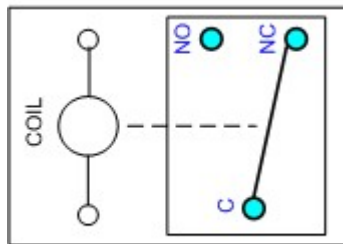


รีเลย์ Relay (สวิตช์ ที่ใช้กระแสไฟฟ้าในสั่งงาน หน้าสัมผัสเปิดปิด)

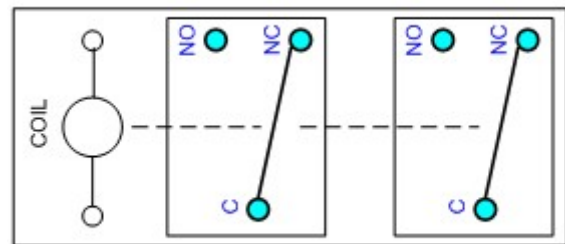




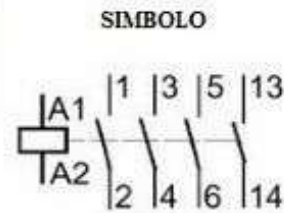
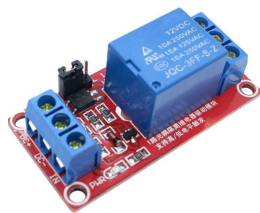
Single Pole, Single Throw (SPST)



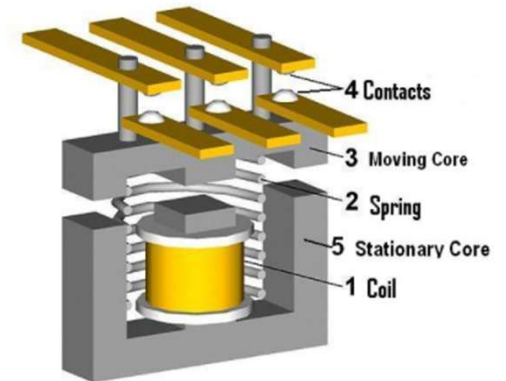
Single Pole, Double Throw (SPDT)



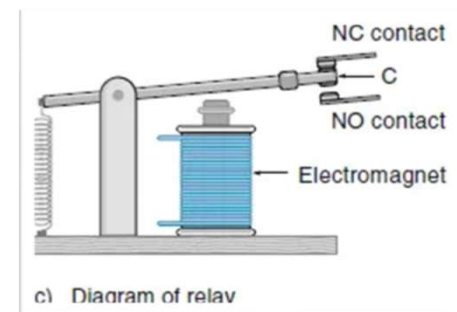
Double Pole, Double Throw (DPDT)



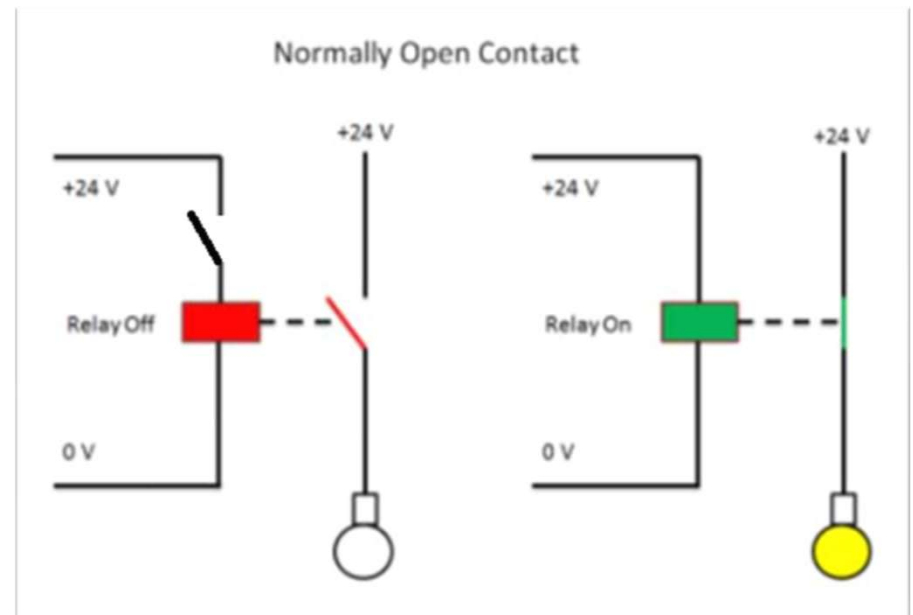
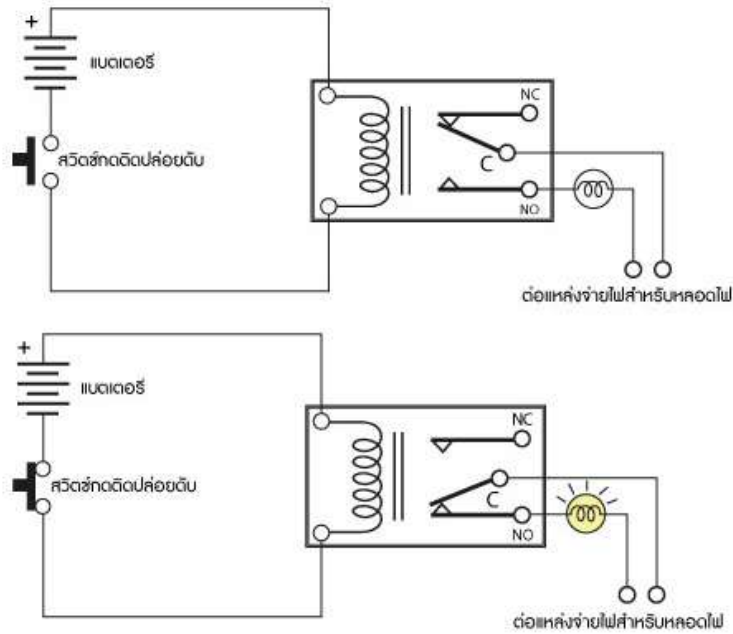
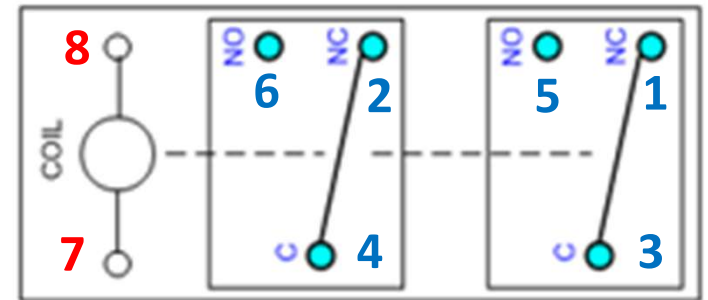
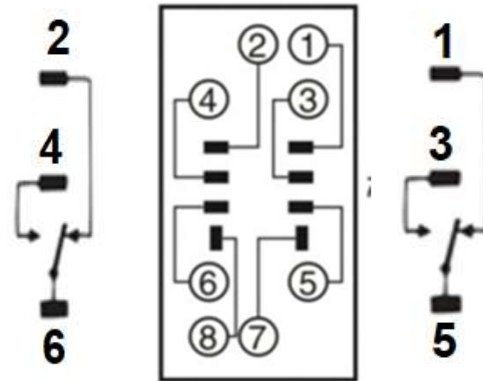
www.areatecnologia.com

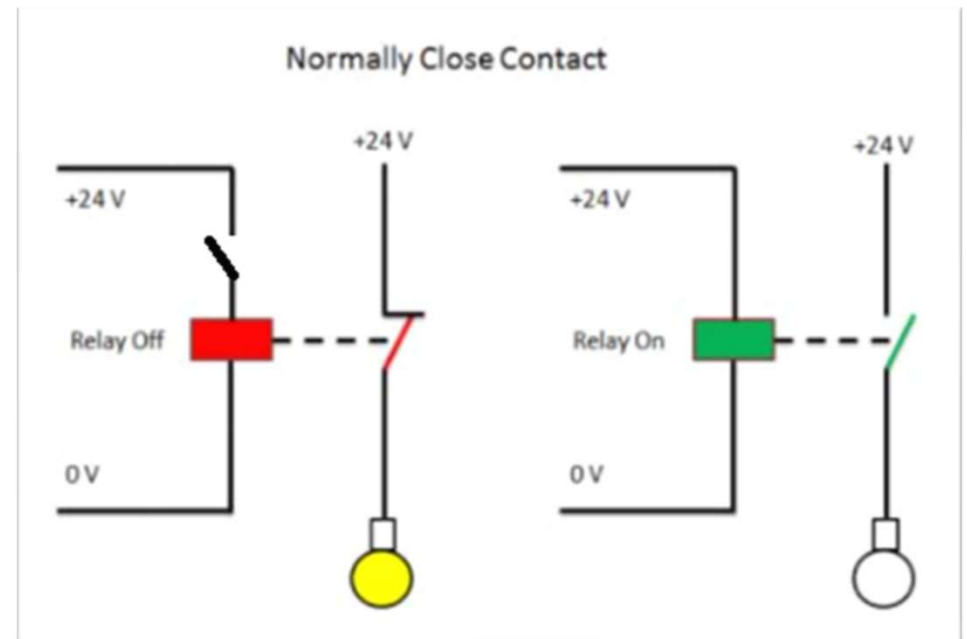
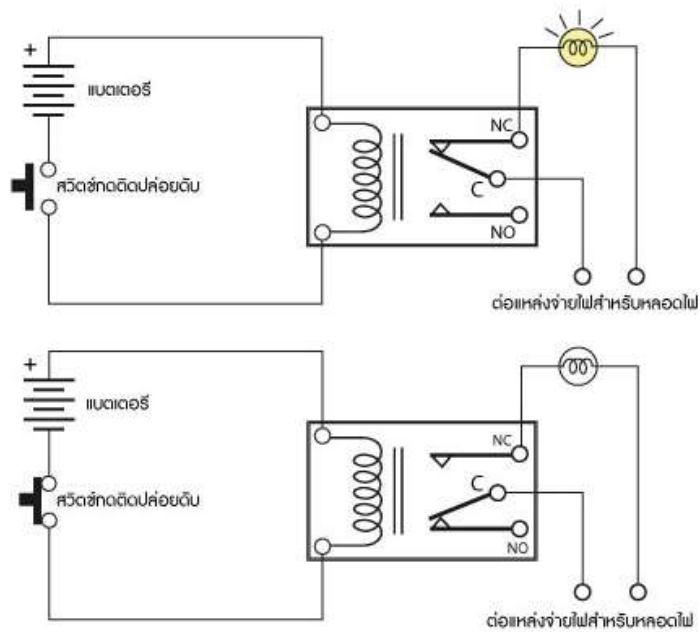
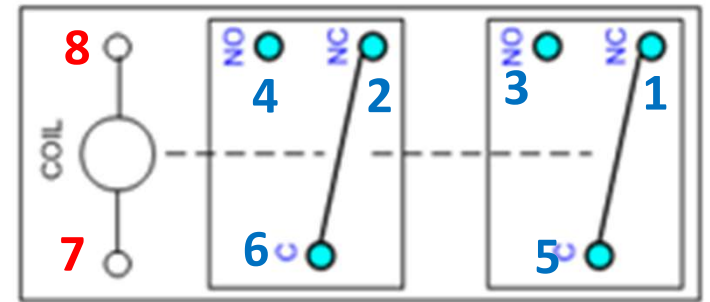
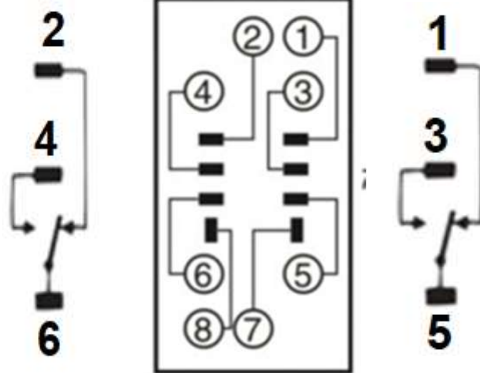


รูปที่ 2

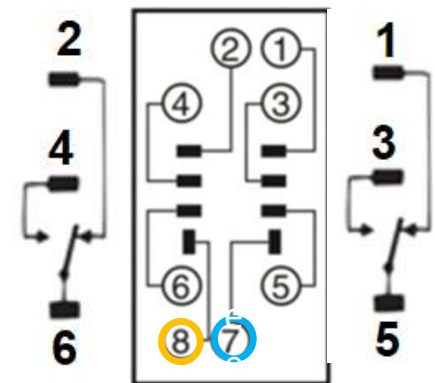
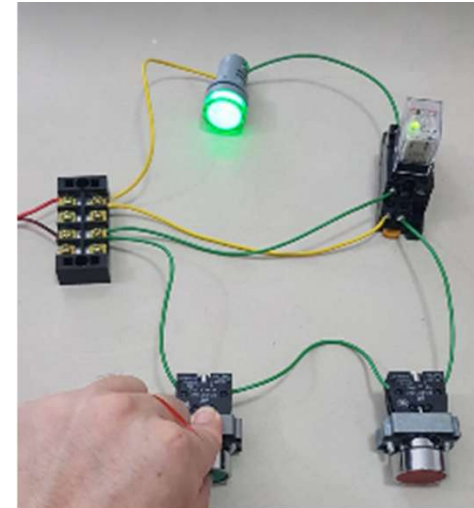
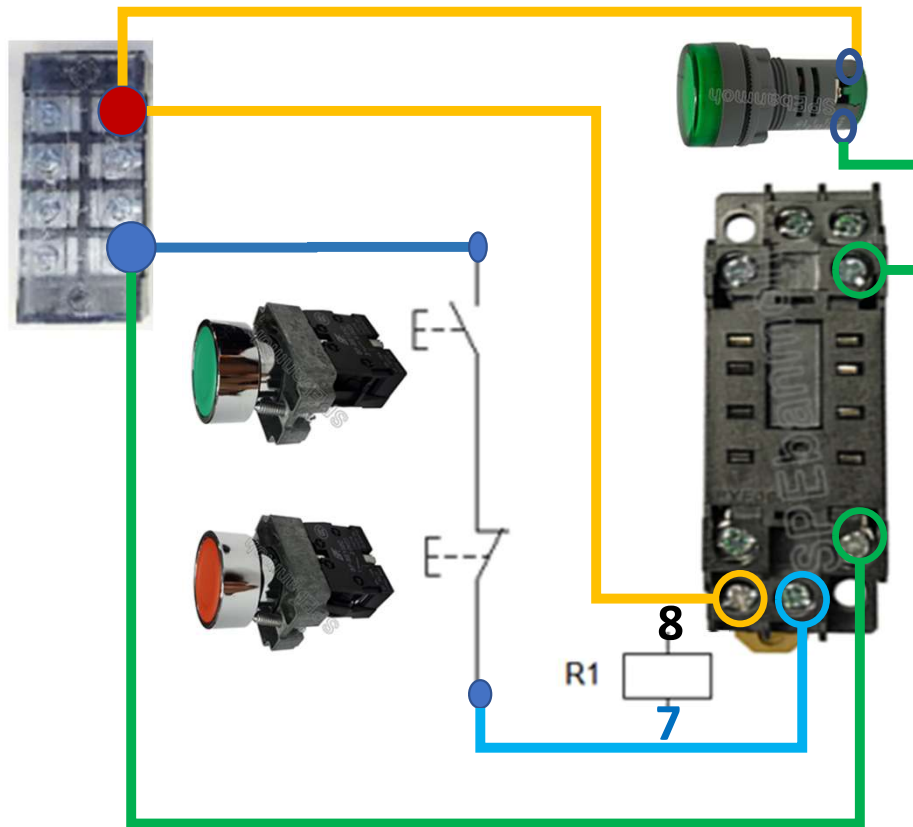
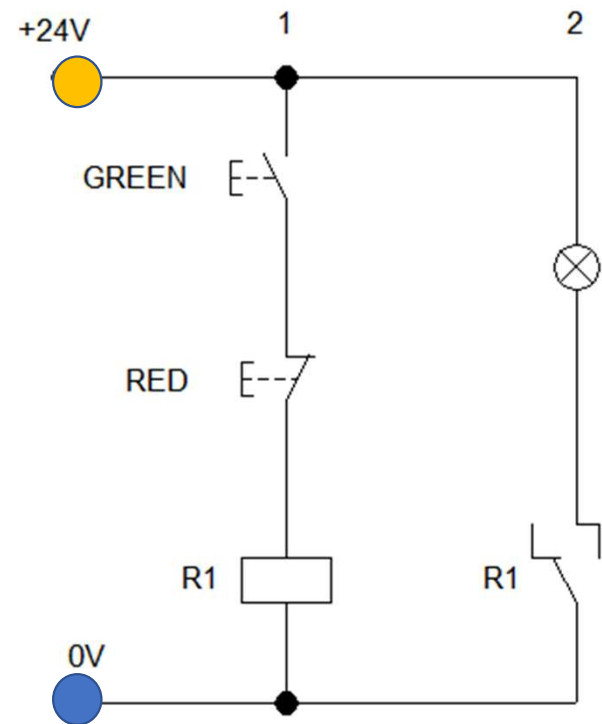
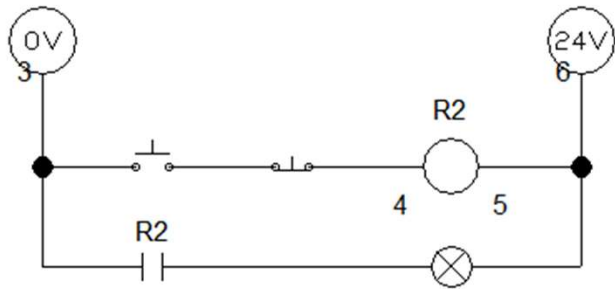


รีเลย์ Relay Switch

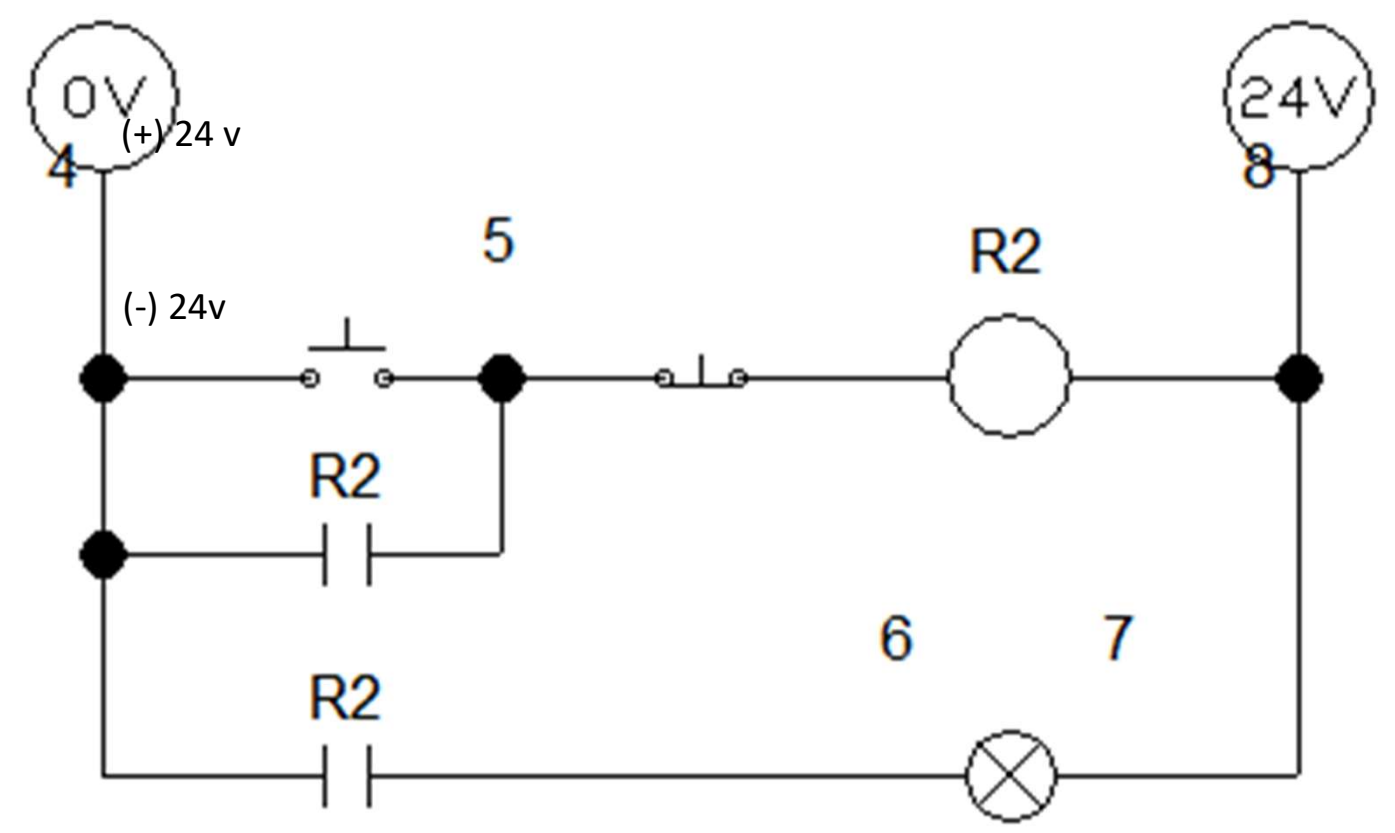
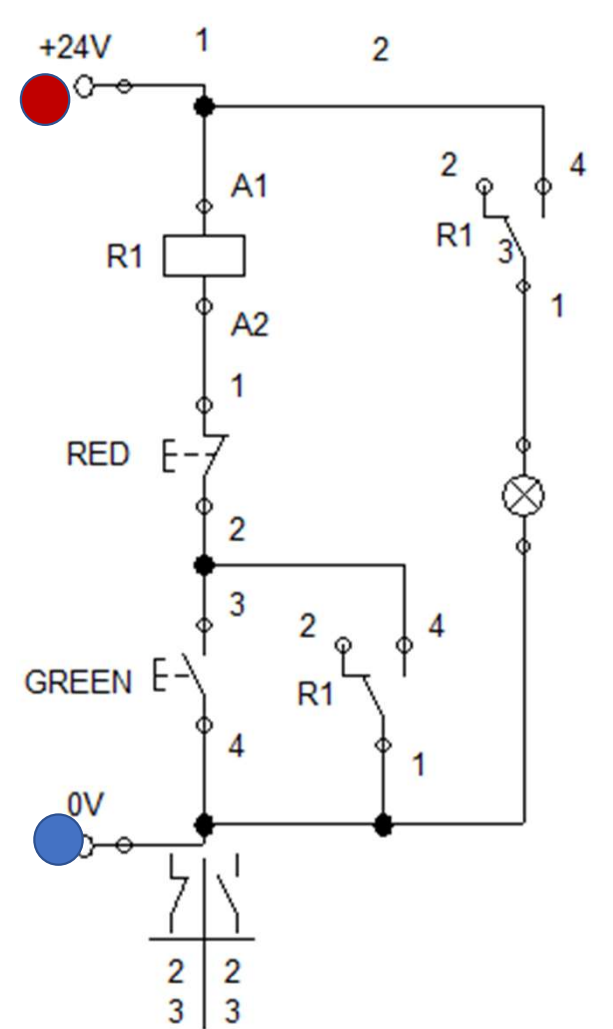


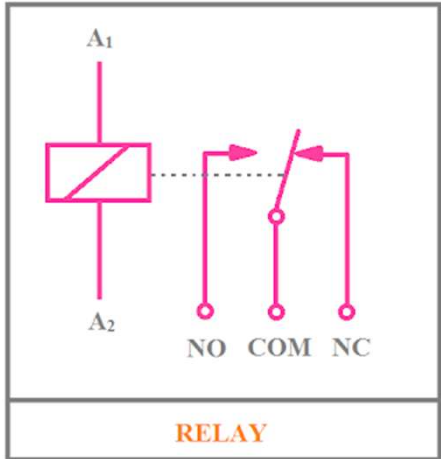


Basic 02 - Relay Control [youtube](#)



Basic 03 - Relay Control (Self Holding) [youtube](#)





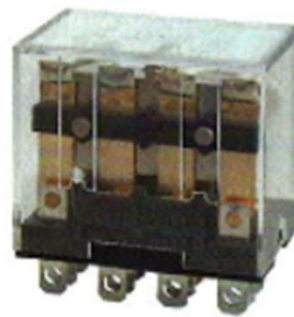
MY2 N
Price:



MY4 N
Price:



LY2N
Price:



LY4N
Price:



MK2P-I
Price:



MK3P-I
Price:



PYF08A
Price:



PYF14A
Price:



PYF08A-E
Price:



PYF14A-E
Price:



PF083A-E
Price:

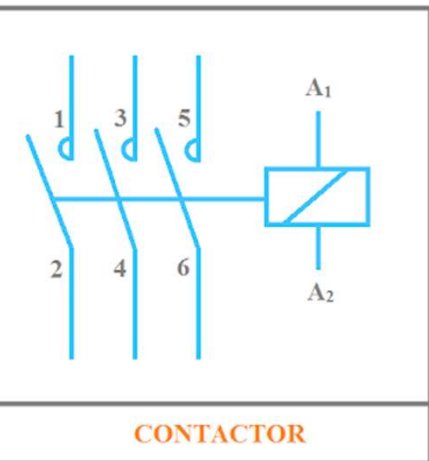


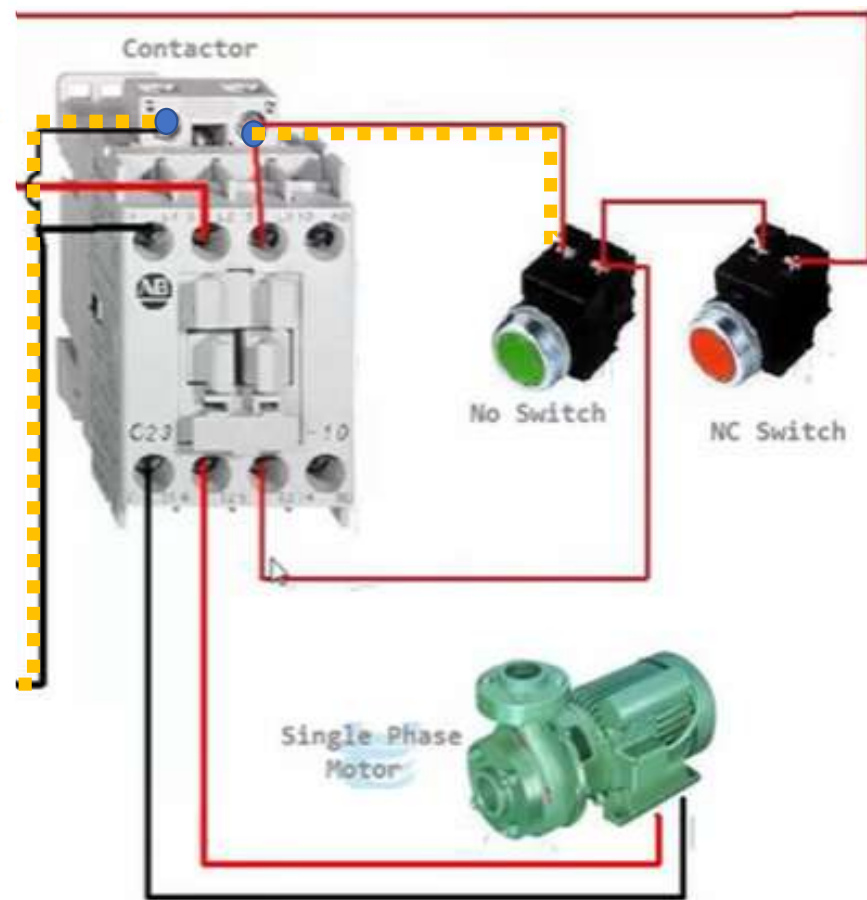
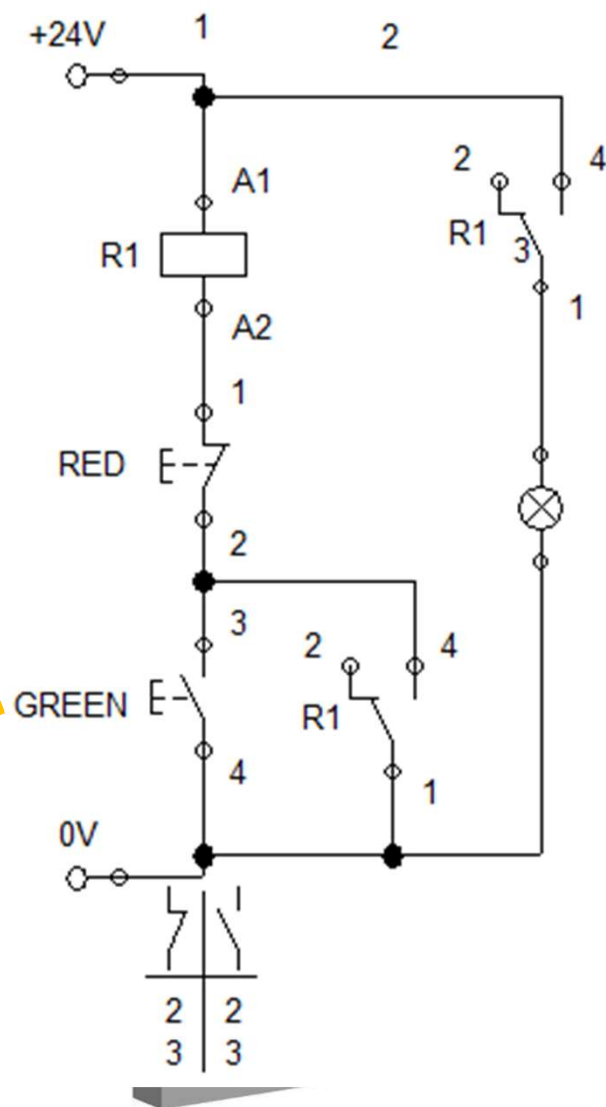
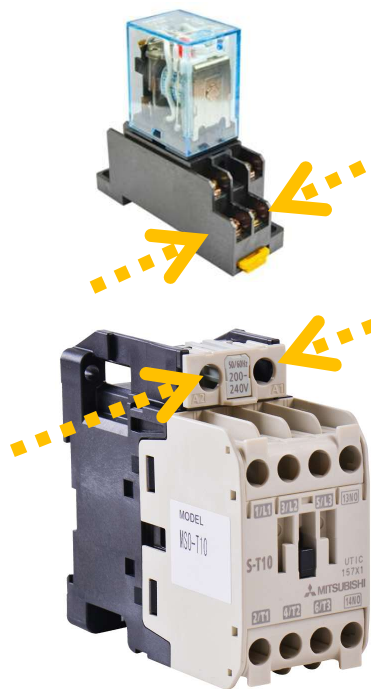
PF113A-E
Price:



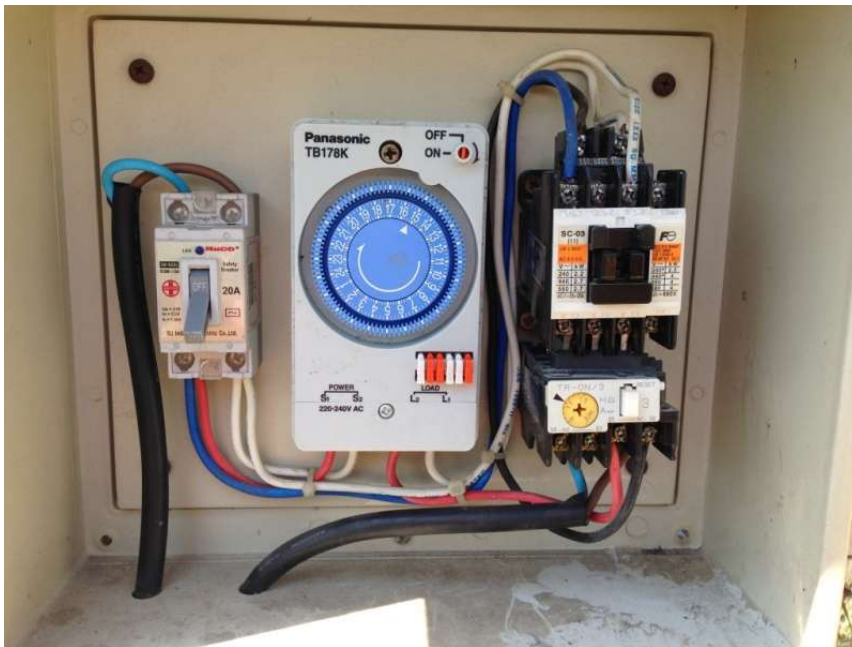
PTF14A-E
Price:

Specifications		Model	MY2 N	MY4 N	LY2	LY4	MK2P-I	MK3P-I
Contact capacity	AC Position		5A 250V	3A 250V	10A 250V	10A 250V	10A 250V	10A 250V
	DC		5A 30V	3A 30V	10A 30V	10A 30V	10A 30V	10A 30V

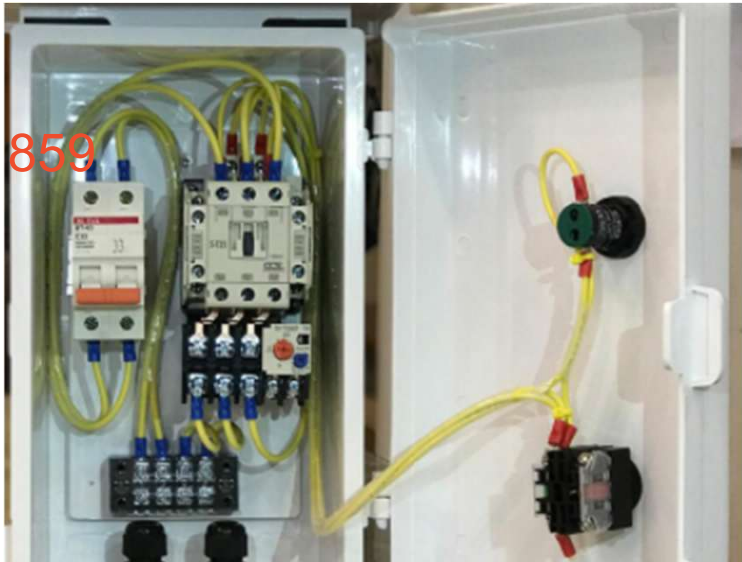




Timer



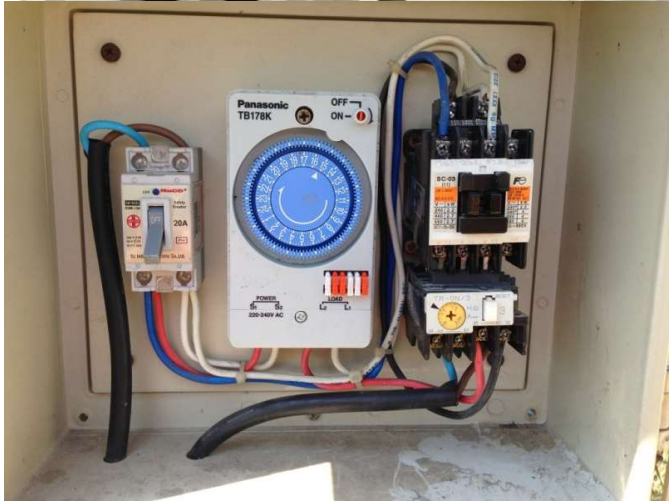
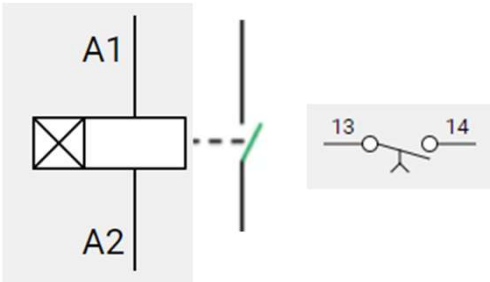
859

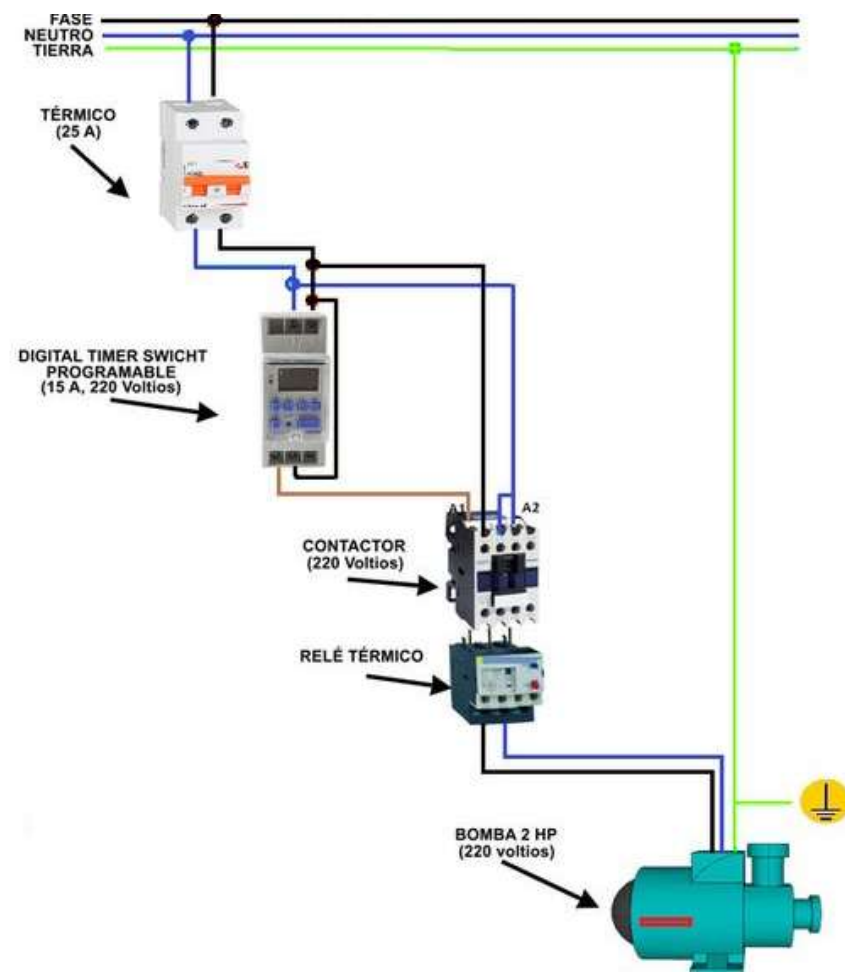
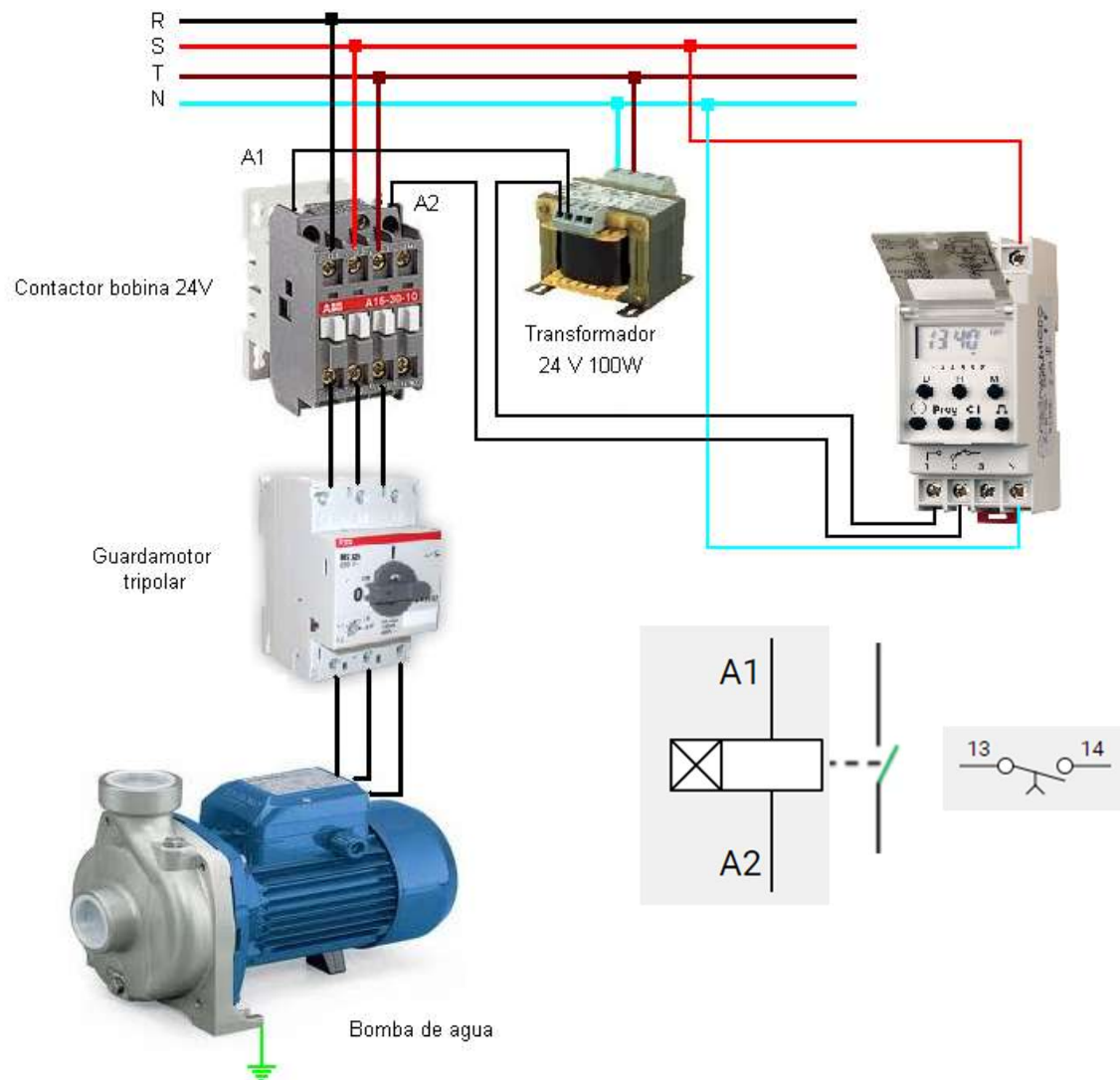


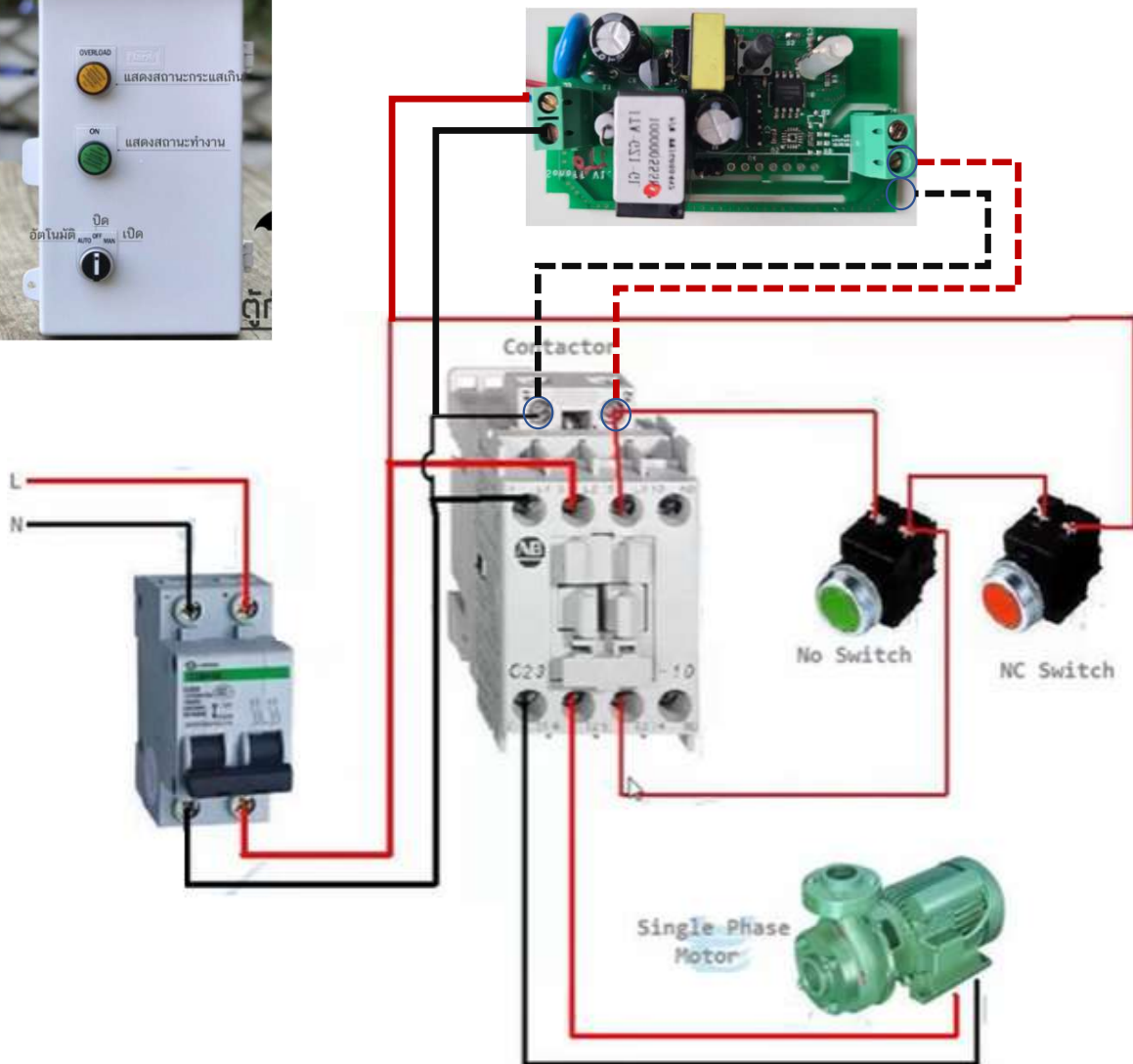
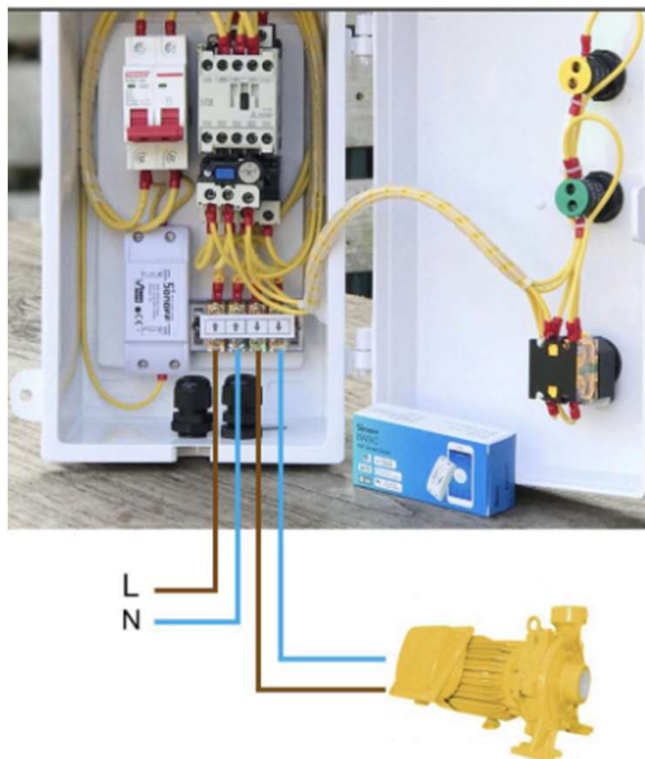
รับประกัน 1 ปี

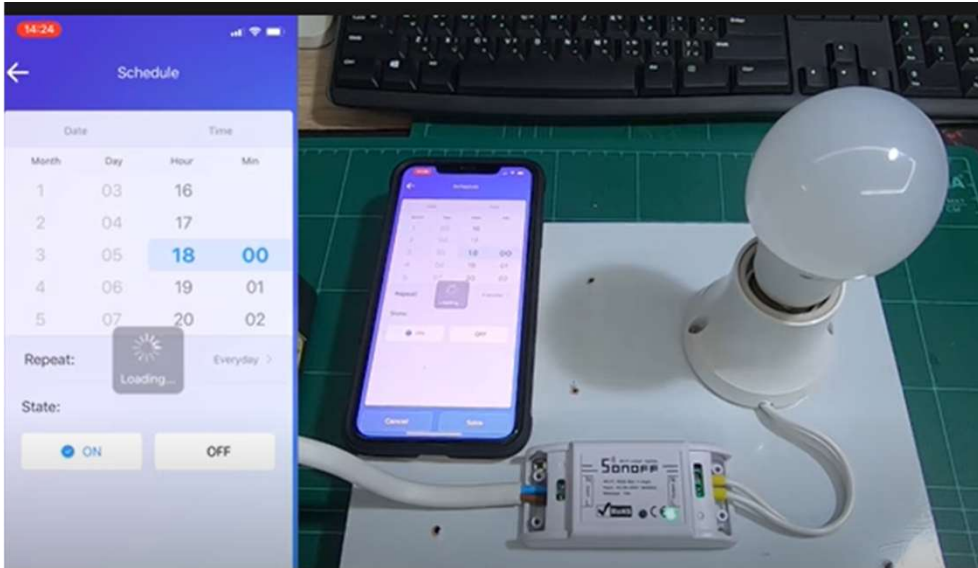
ตัวตั้งเวลา

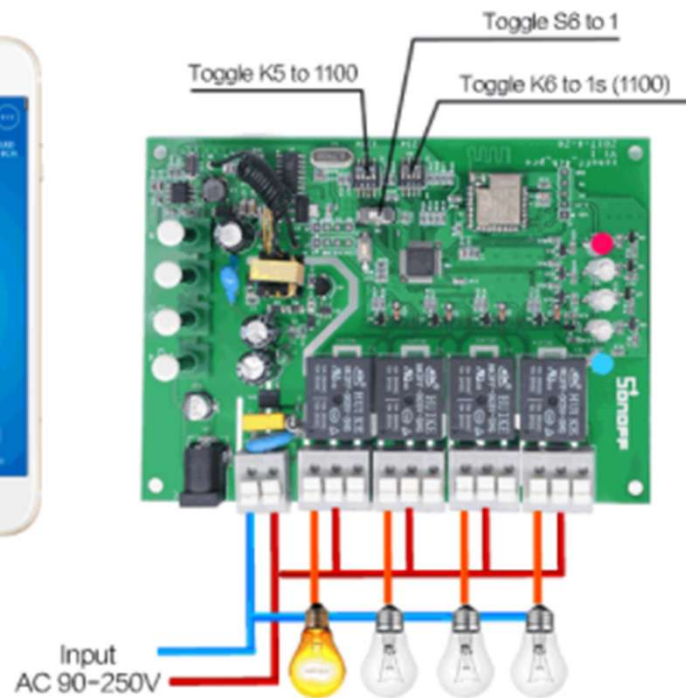
Timer - 1,495











Sonoff 4CH PRO R3
สวิตช์ผ่าน WiFi 4 ช่อง พร้อม รีโมท RF

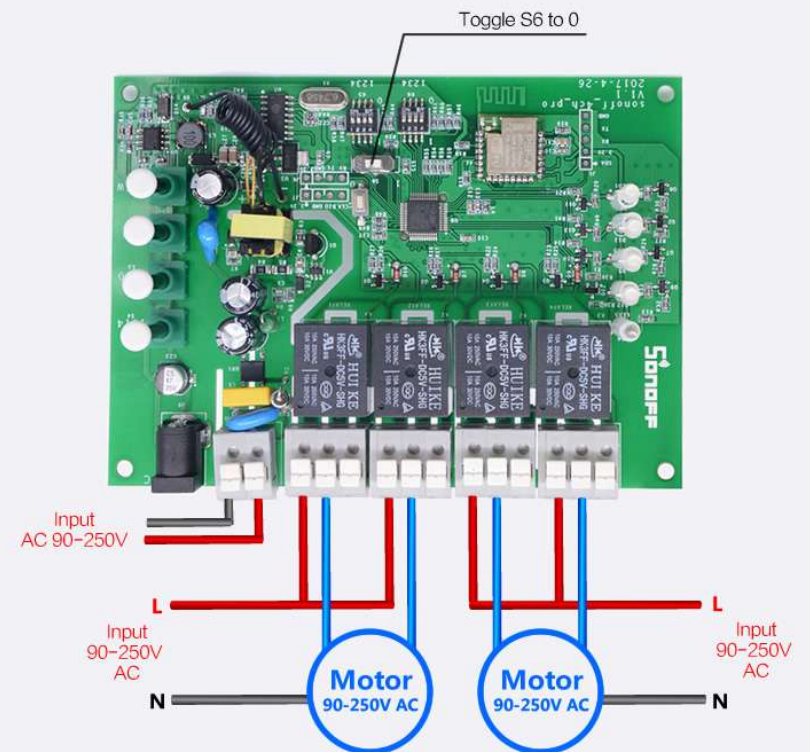
★★★★★ 78 รีวิว | สำรอง 46 รายการ

แบรนด์: Sonoff | เพิ่มสินค้าที่คุณสนใจในตะกร้า Sonoff in TH

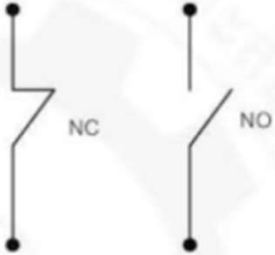
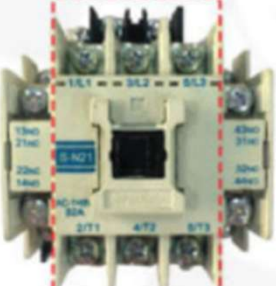
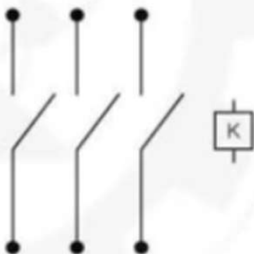
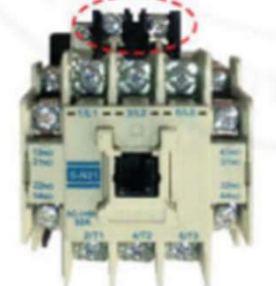
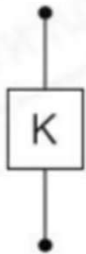
B1,000.00
฿1,200.00 -17%

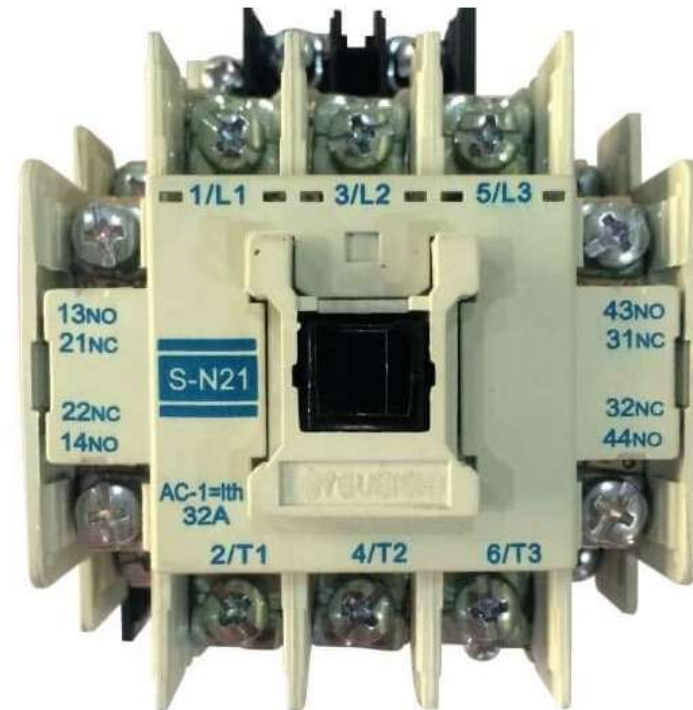
จำนวน 1

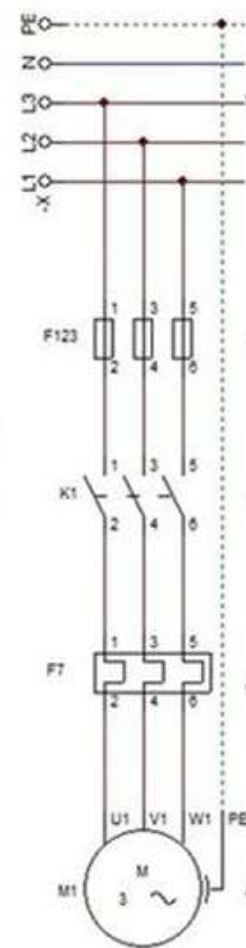
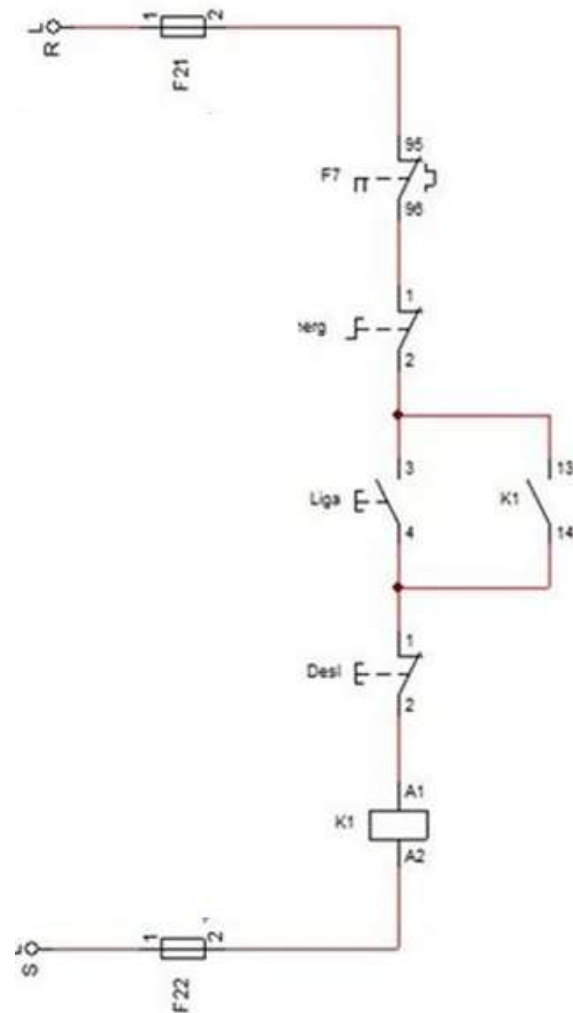
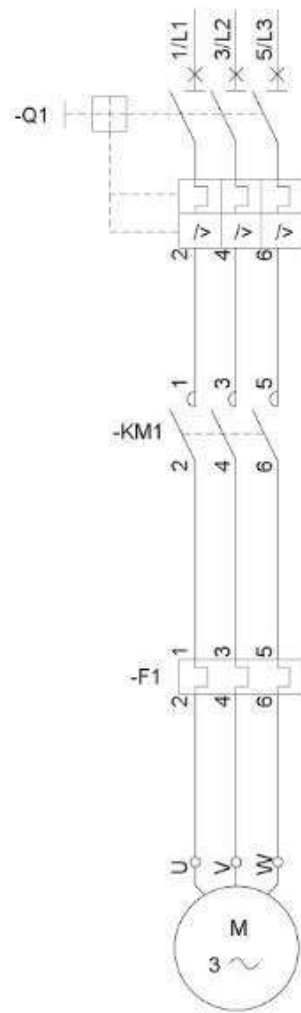
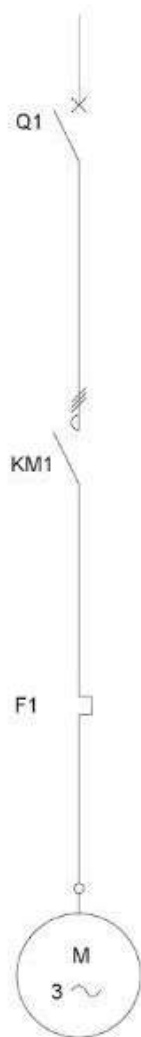
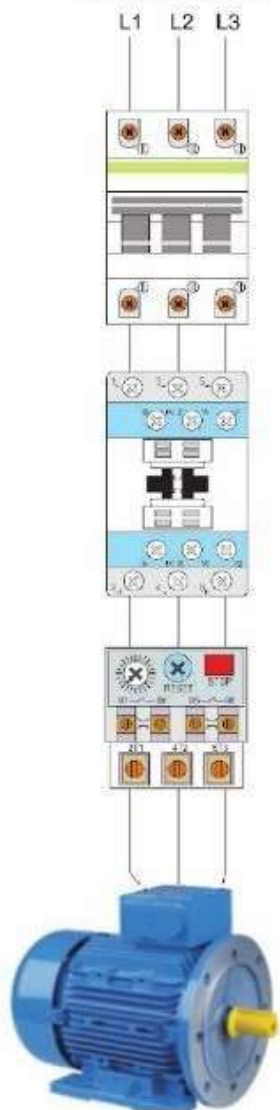
AC Motor Wiring Diagram

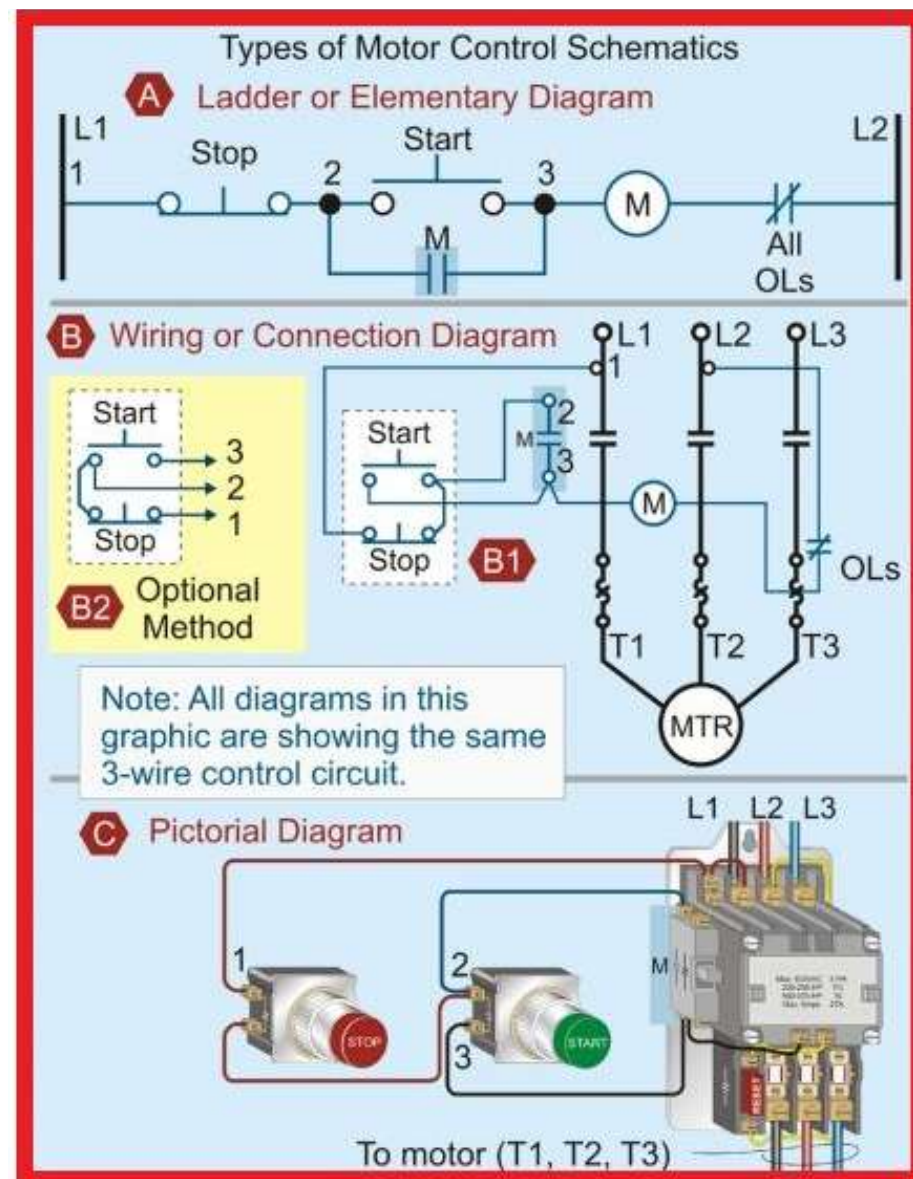
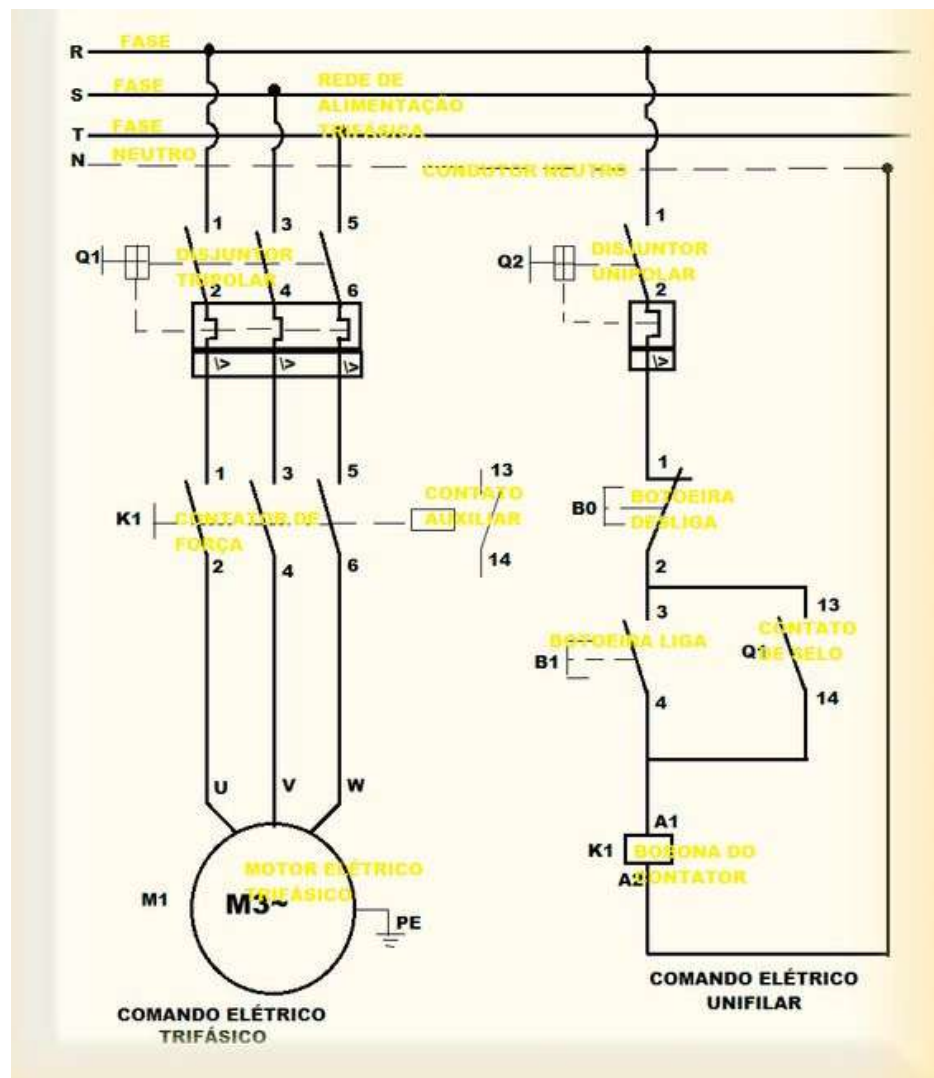


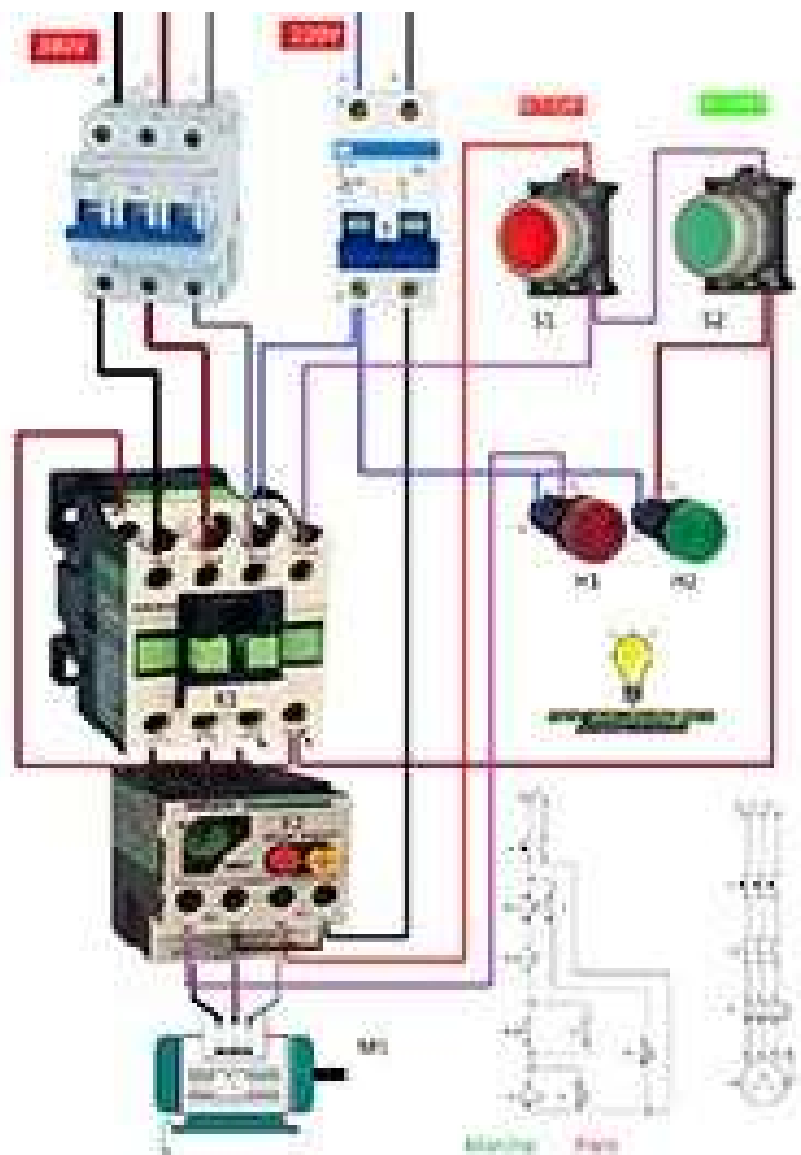
Note : only 1 motor can run each time.

ลักษณะ	วงจรไฟฟ้า
	
	
	

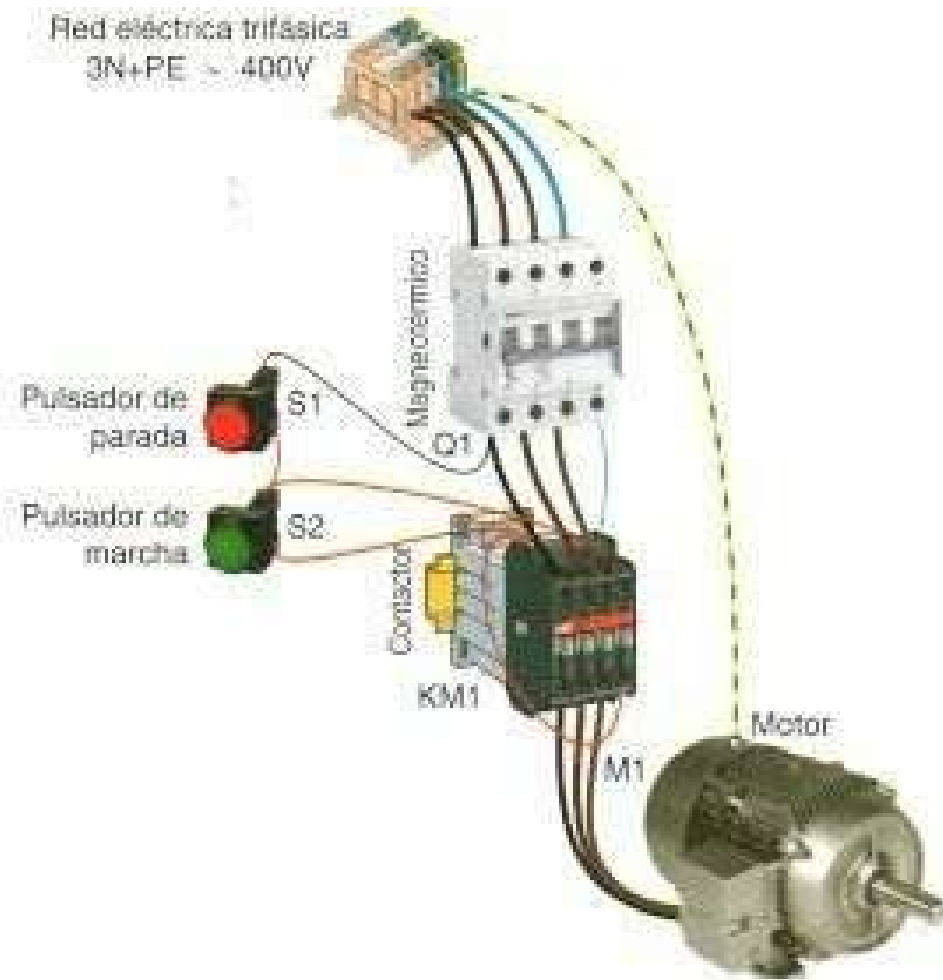


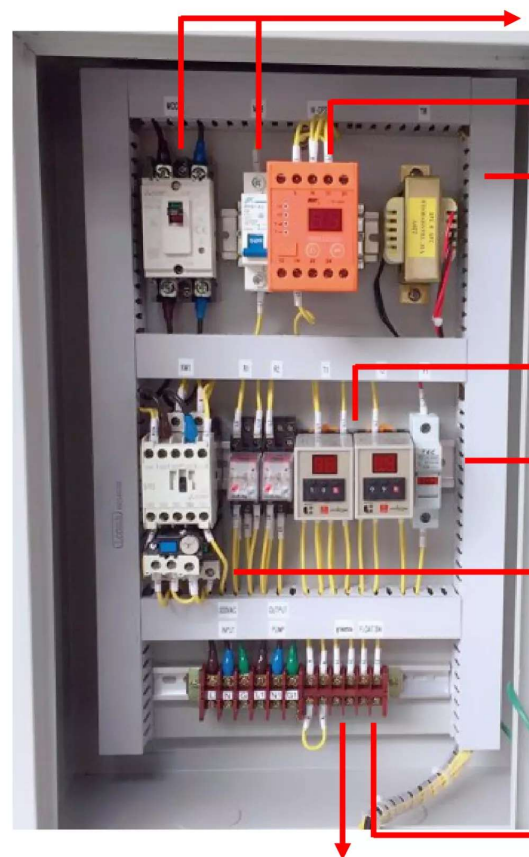
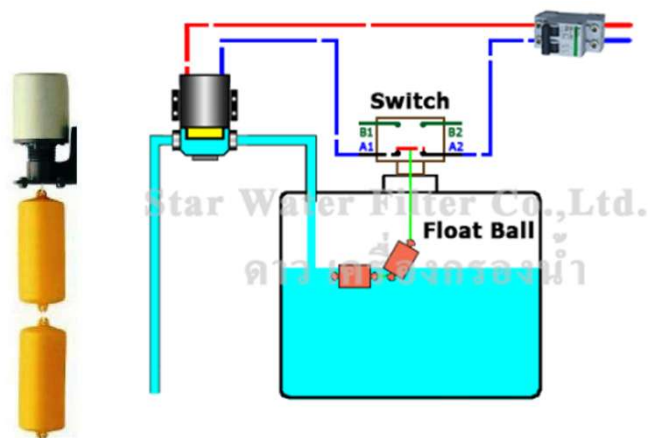
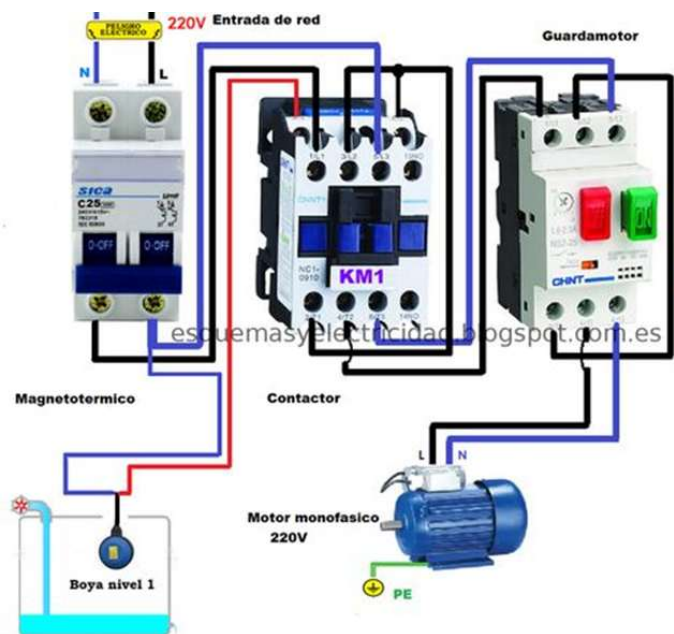






Red eléctrica trifásica
3N+PE ~ 400V





- มี MCCB , MCB ป้องกันการลัดวงจร
- มีระบบป้องกันไฟตกไฟเกิน เพื่อไม่ให้อุปกรณ์ภายในตู้เสียหายได้ รวมถึงปั๊มด้วย
- มีระบบหม้อแปลงไฟจาก 220vac เป็น 24vac เพื่อความปลอดภัยต่อผู้ใช้งาน

- มี Timer หน่วงเวลาเช็คระบบน้ำในเส้นท่อ แล้วกลับมาทำงานของปั๊มน้ำแบบอัตโนมัติ

- มี Fuse control ป้องกันการลัดวงจรแรงดันต่ำ

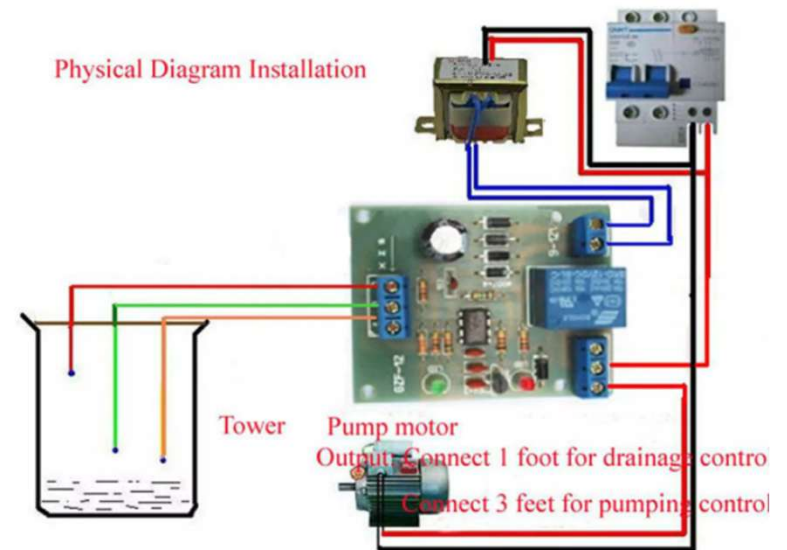
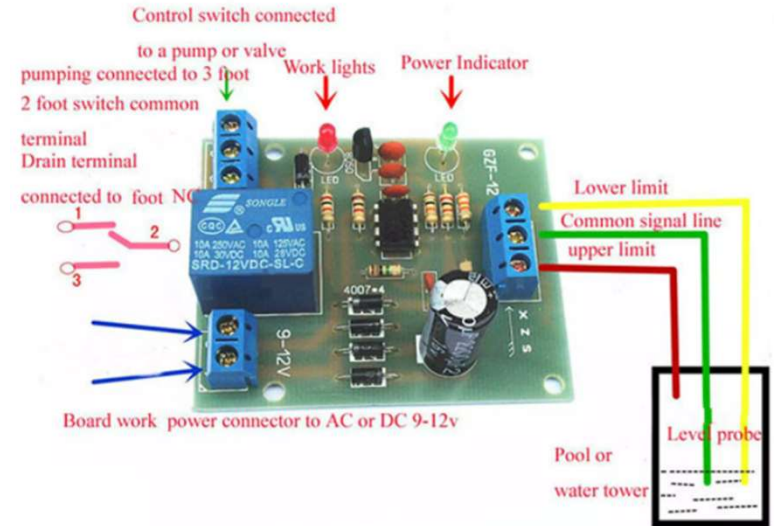
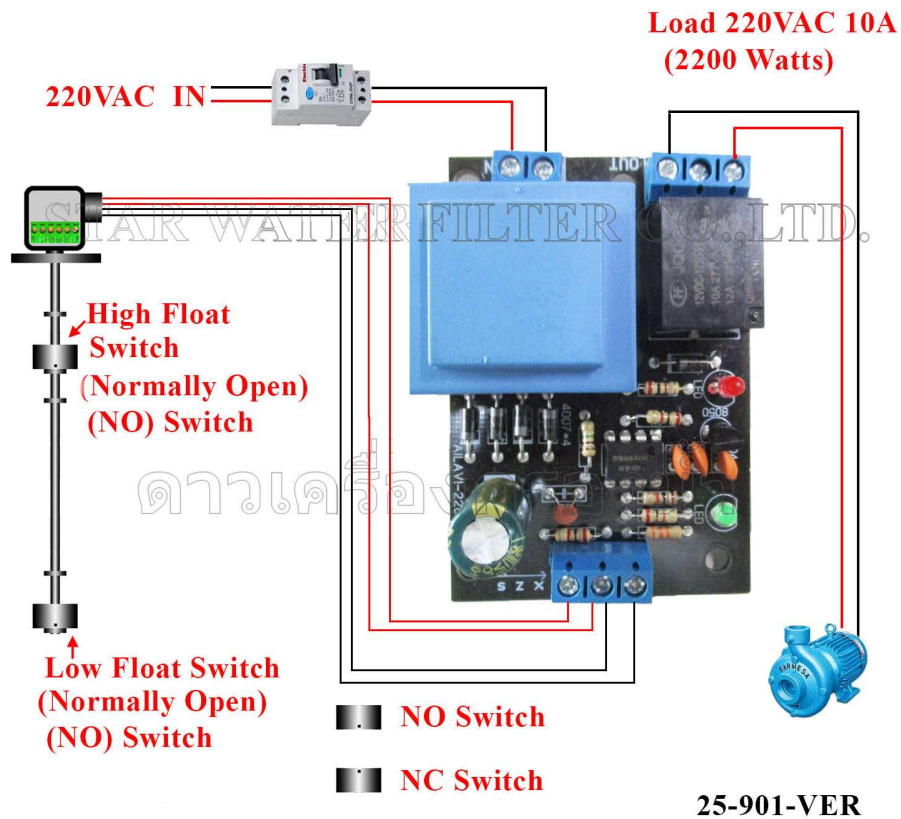
- มีโอเวอร์โวลต์รีเลย์ ป้องกัน การกินเกินกระแสมอเตอร์

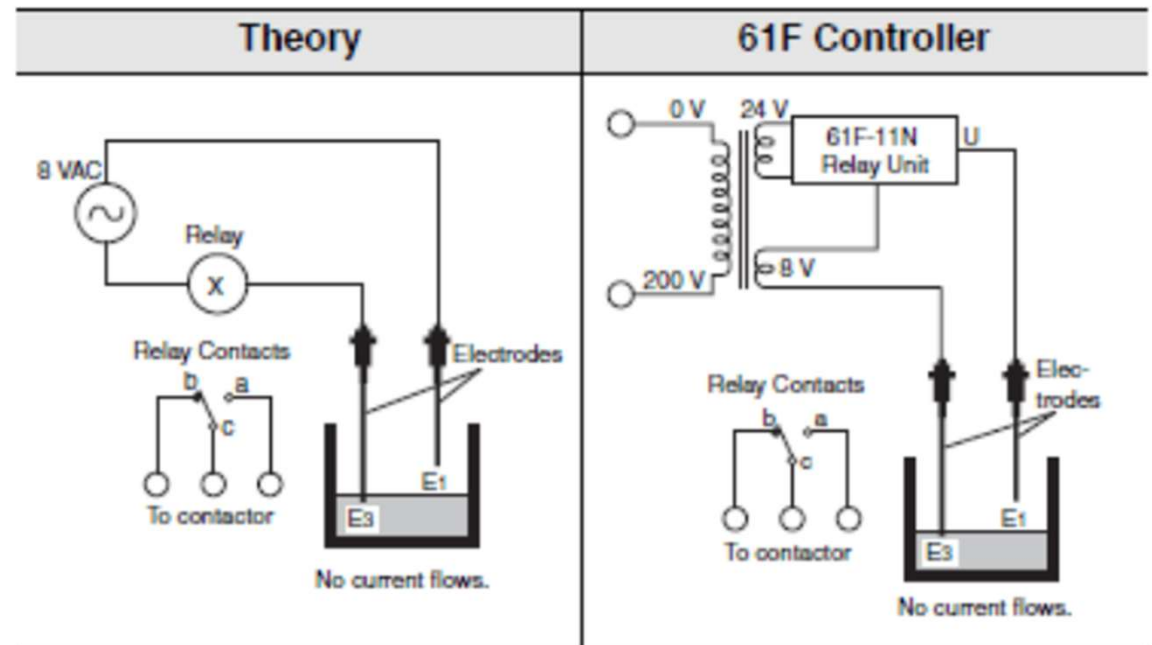
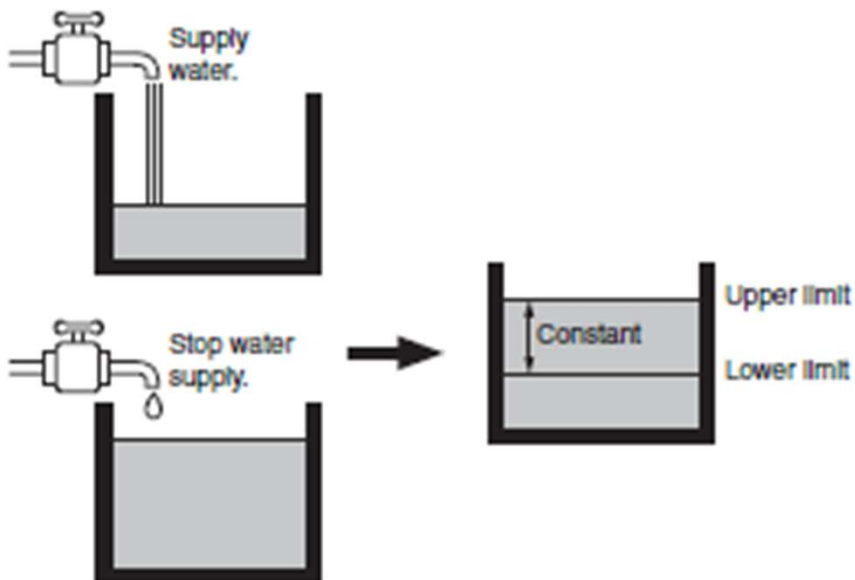


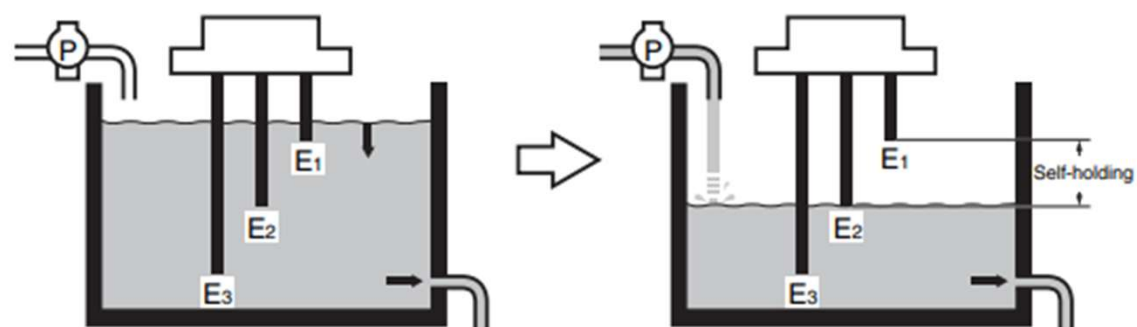
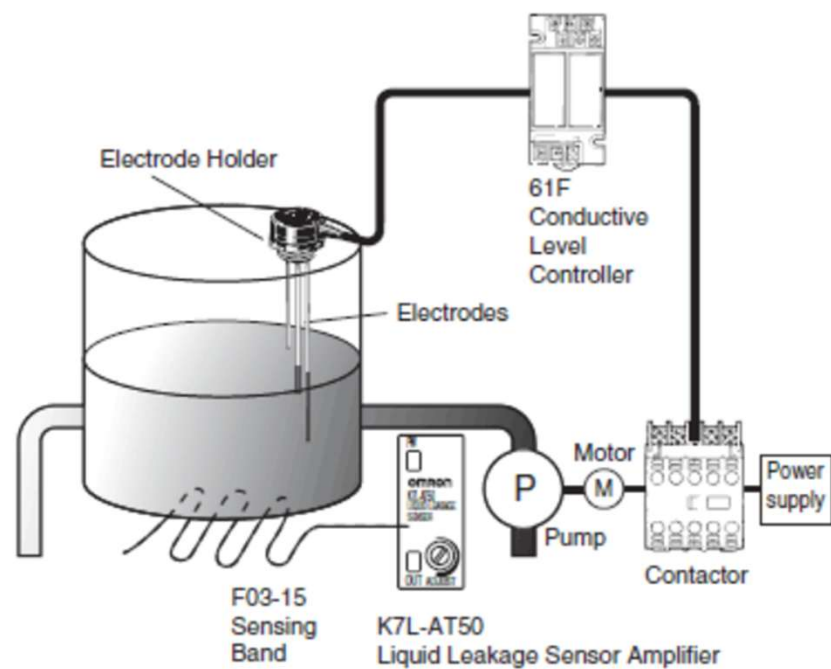
- มีระบบป้องกันปั๊มน้ำ Run dry เพื่อไม่ให้ปั๊มน้ำทำงานตัวเปล่า อาจทำให้ปั๊มไหม้เสียหายได้ ด้วย โฟลสวิทช์

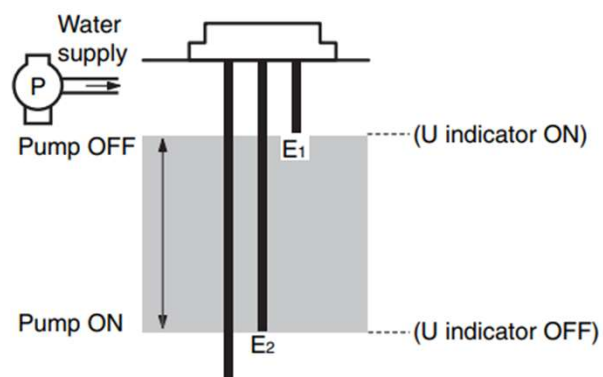
Float Switch

High Low Water Level Controller Module 10A 220VAC





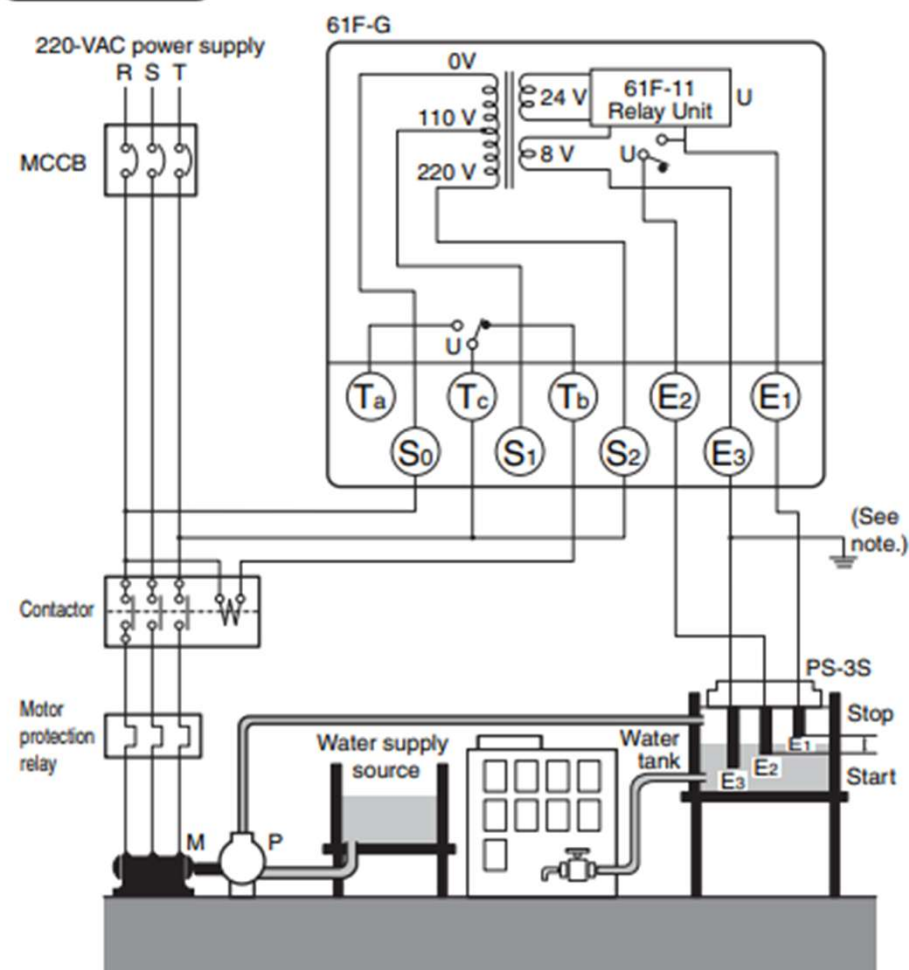




- Connect Tb to the contactor's coil terminal.
- Power Supply Connections (for models with 110/220-V power)
110 VAC: Connect S₀ and S₁.
220 VAC: Connect S₀ and S₂.

Automatic Water Supply Control

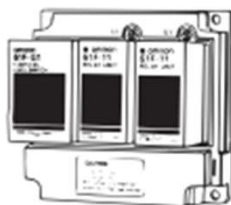
Connections



Note: Be sure to ground the common Electrode (the longest Electrode).

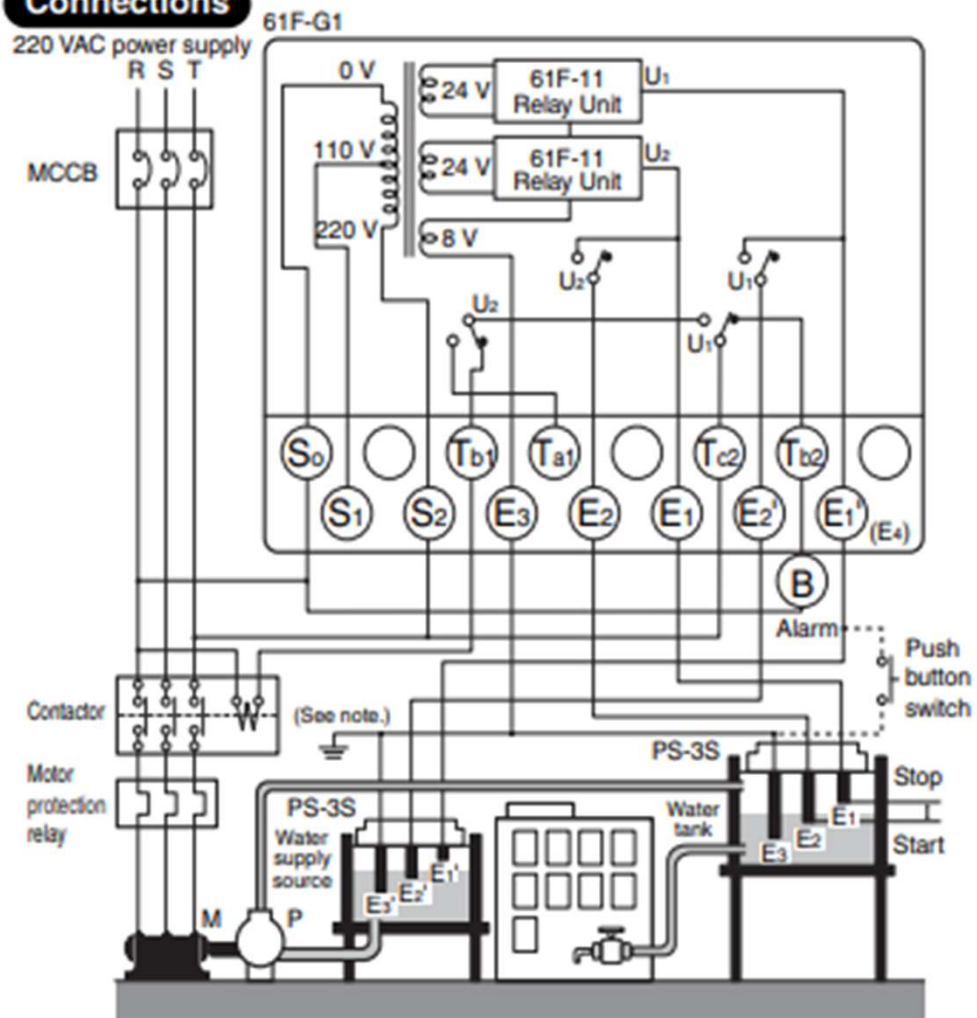
61F-G1

Dimensions:
page 14



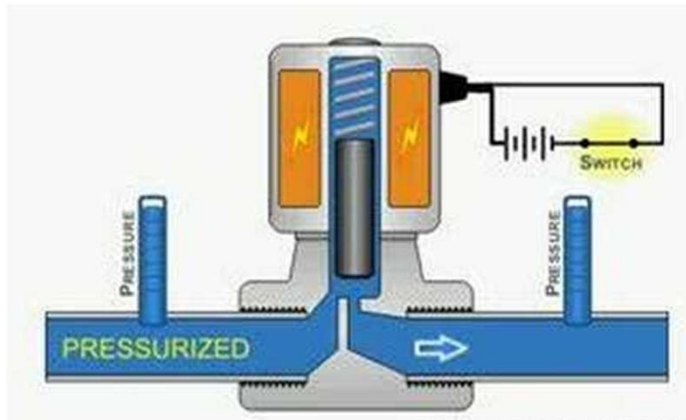
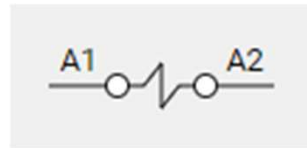
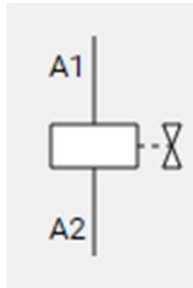
Connections

220 VAC power supply

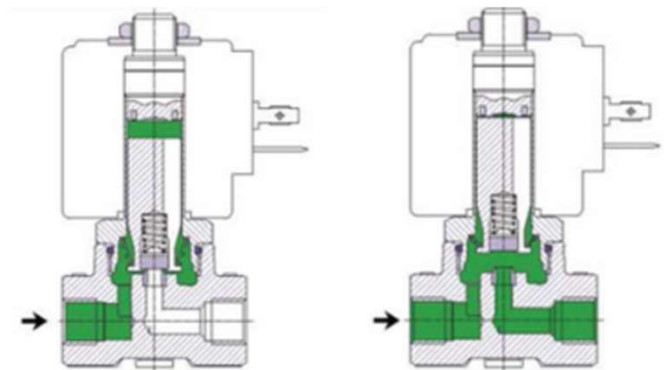
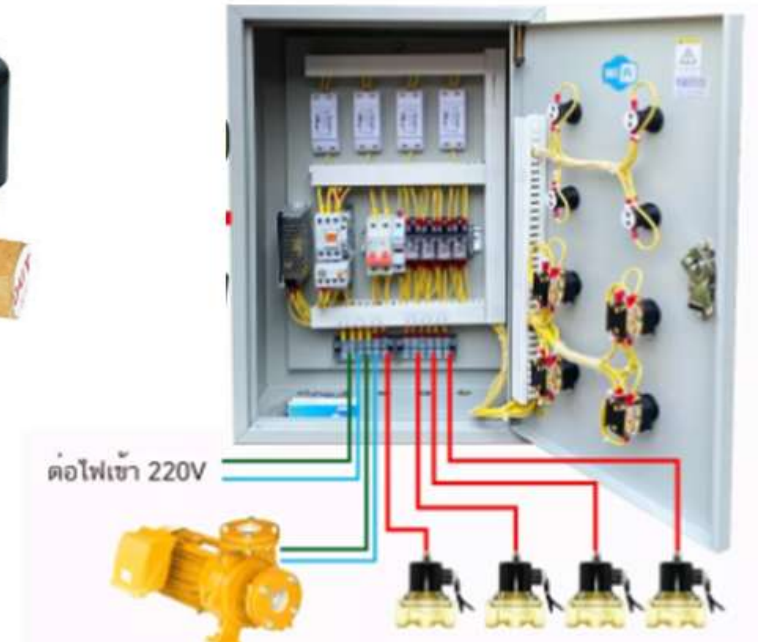


Note: Be sure to ground the common Electrode (the longest Electrode).

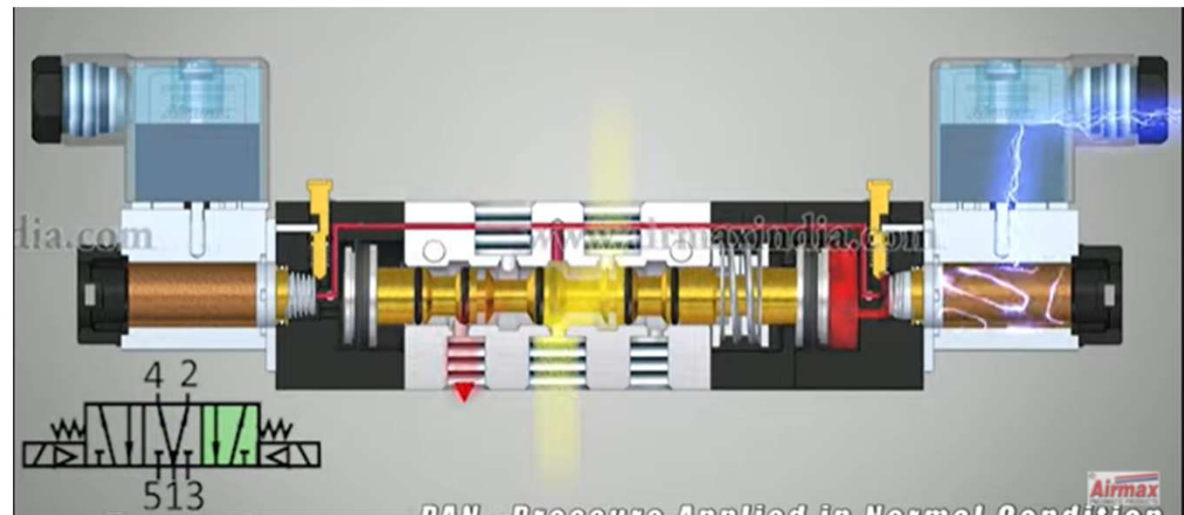
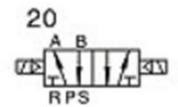
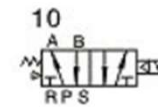
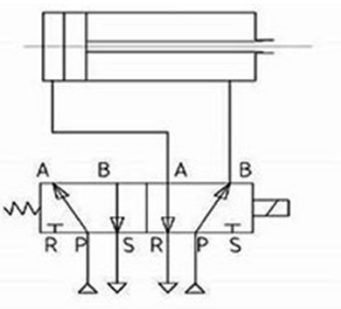
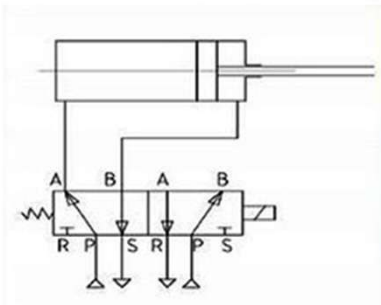
Solenoid Valves



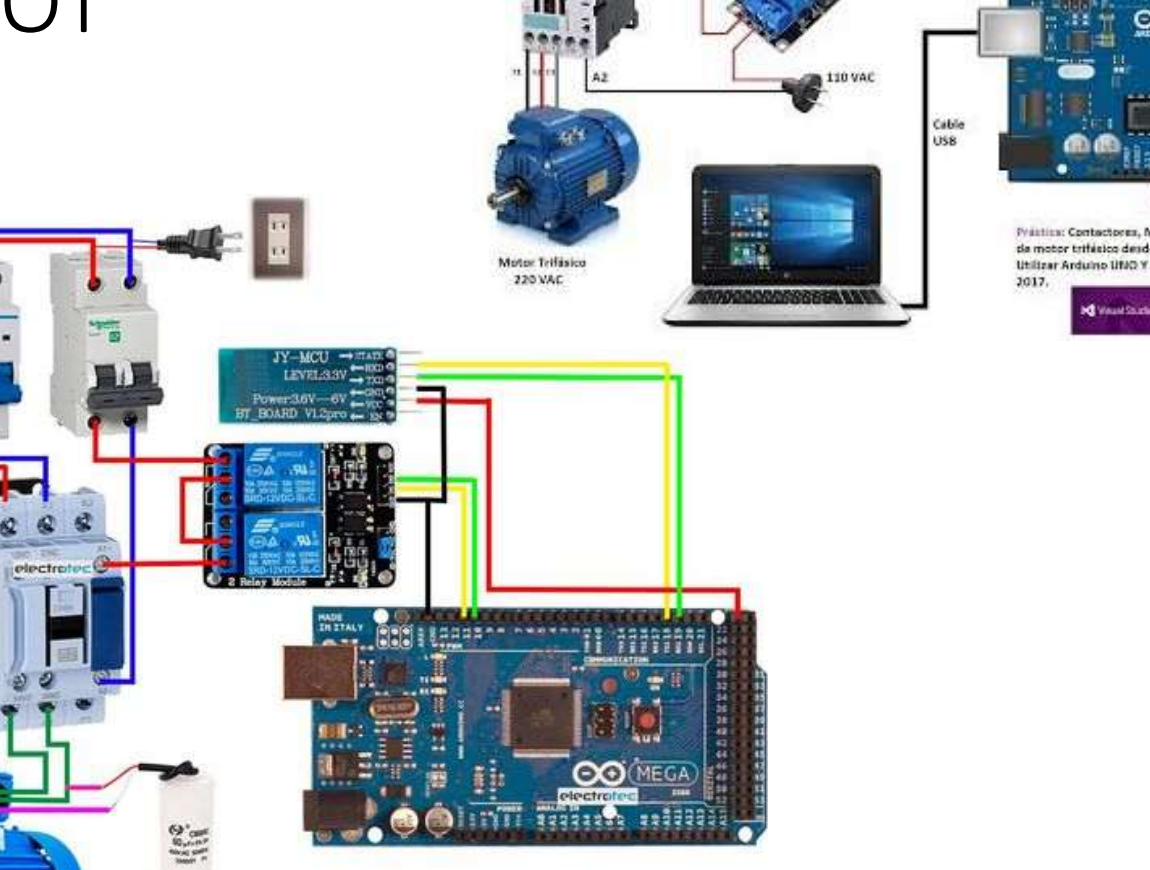
ระบบรดน้ำอัตโนมัติ 4 โซน



Solenoid Valves



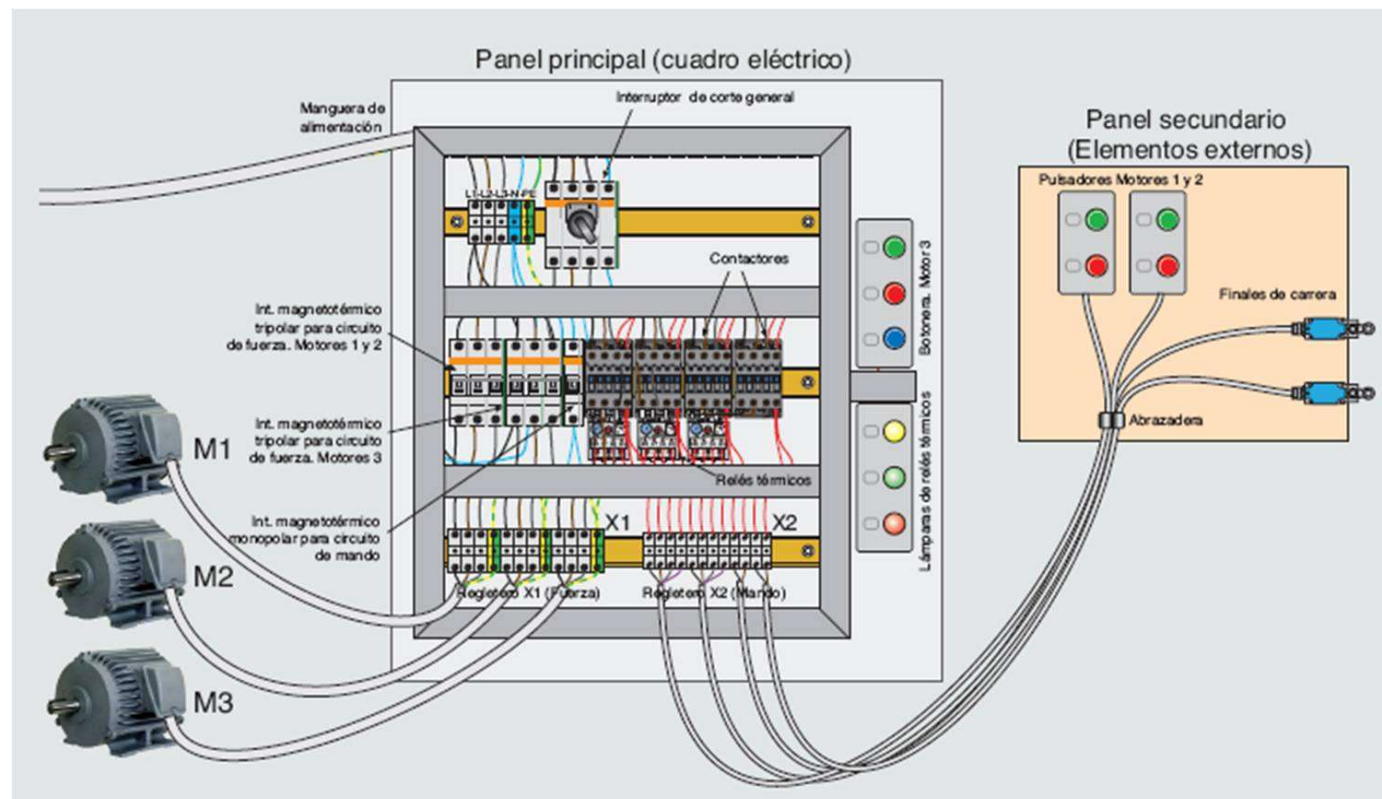
IOT

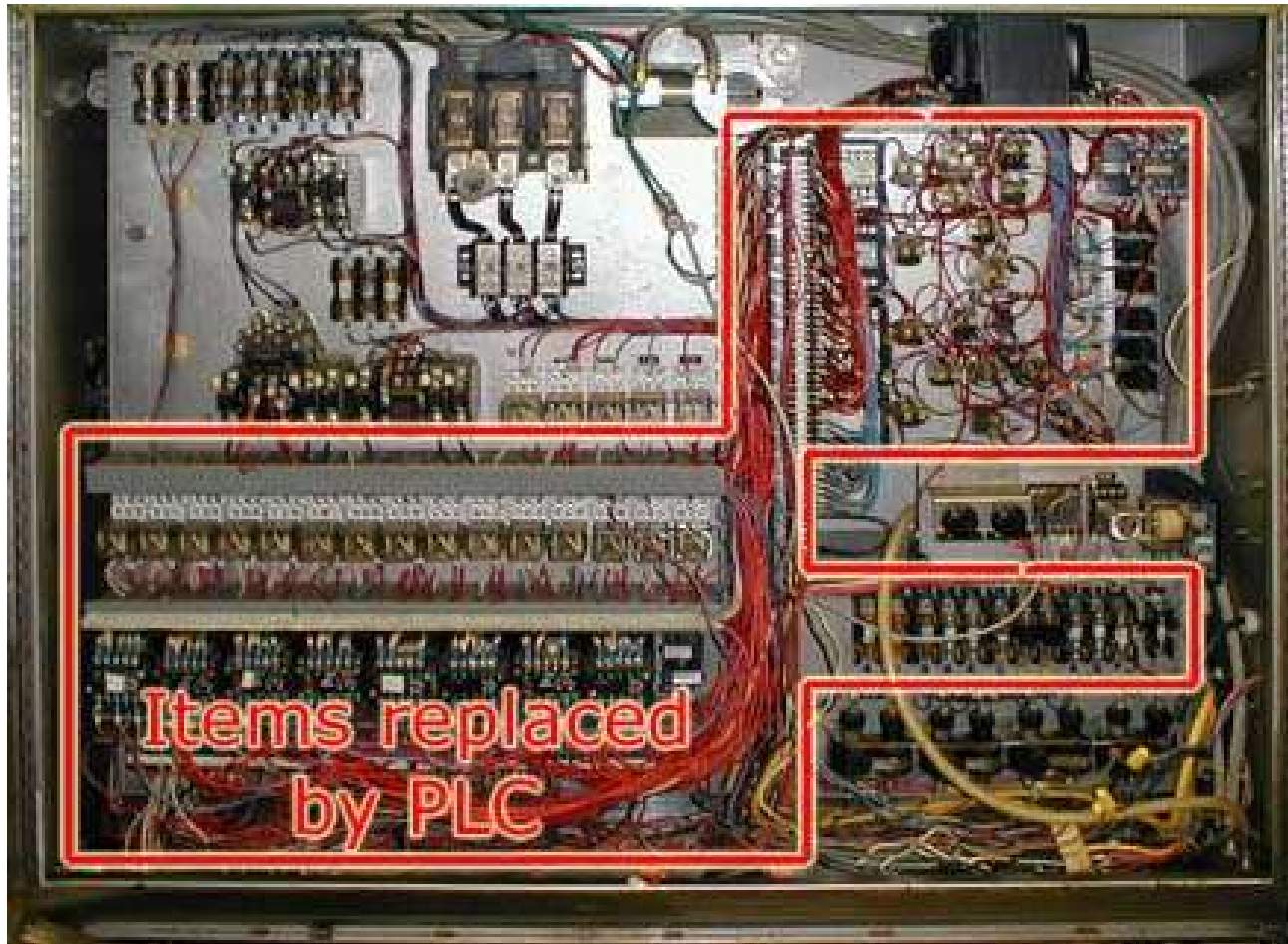


Práctica: Contactores, Marcha y Paro de motor trifásico desde una Laptop. Utilizar Arduino UNO Y Visual Studio 2017.

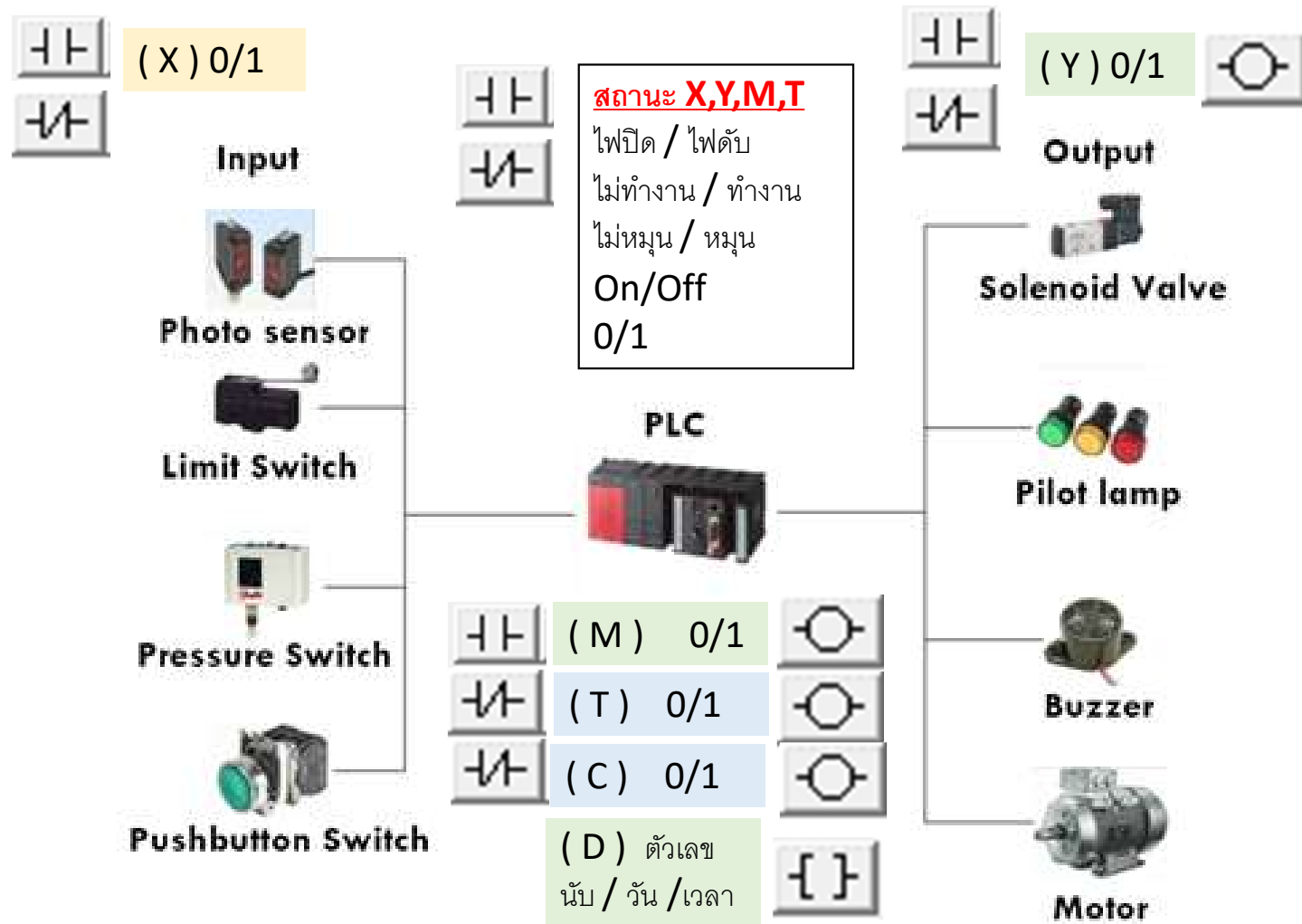
Visual Studio 2017







PLC (Programable Logic Control)



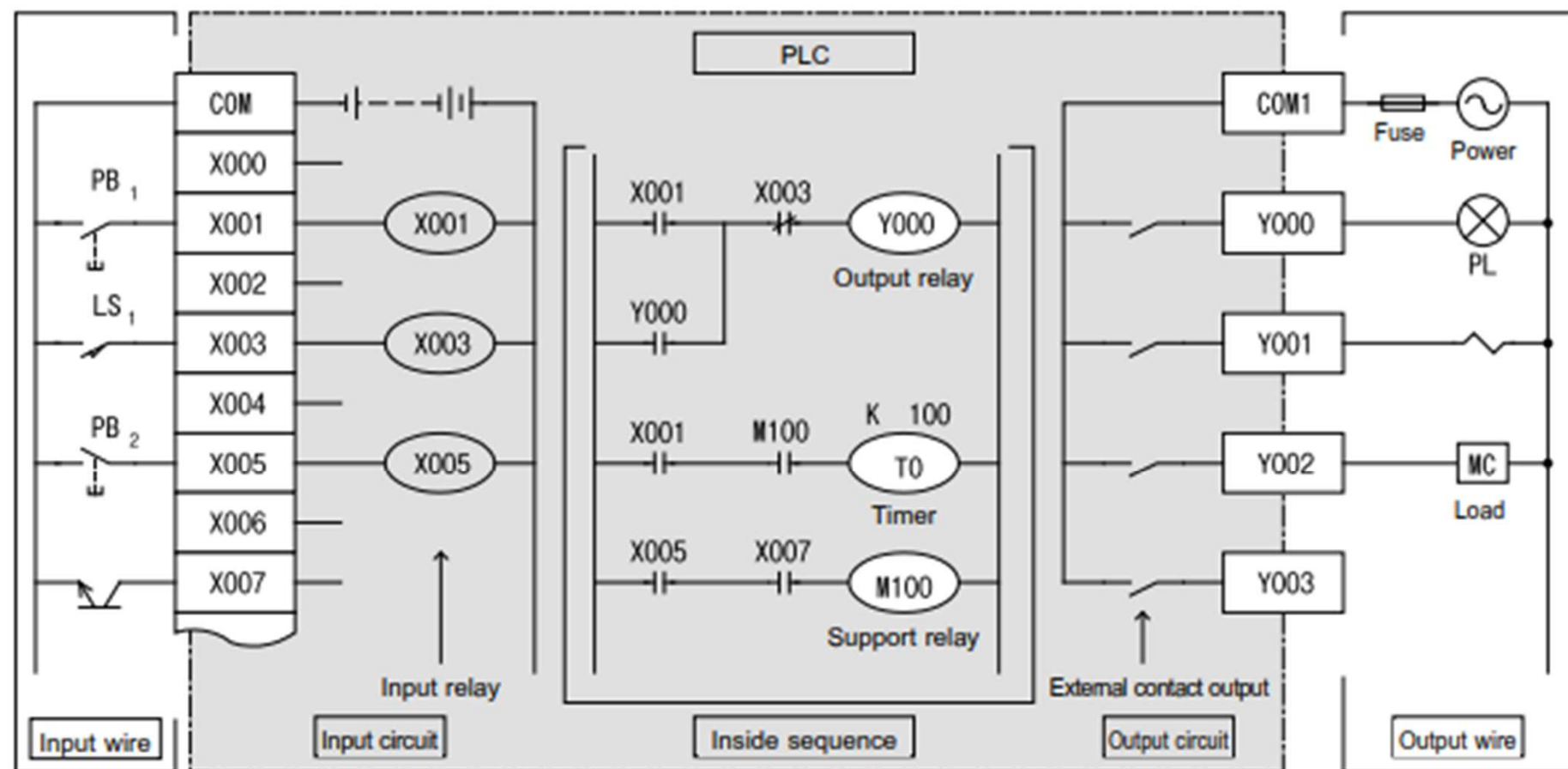
- (X) Input
- (Y) Output
- (M) Mem Relay
- (T) Timer
- (C) Counter
- (D) Memory

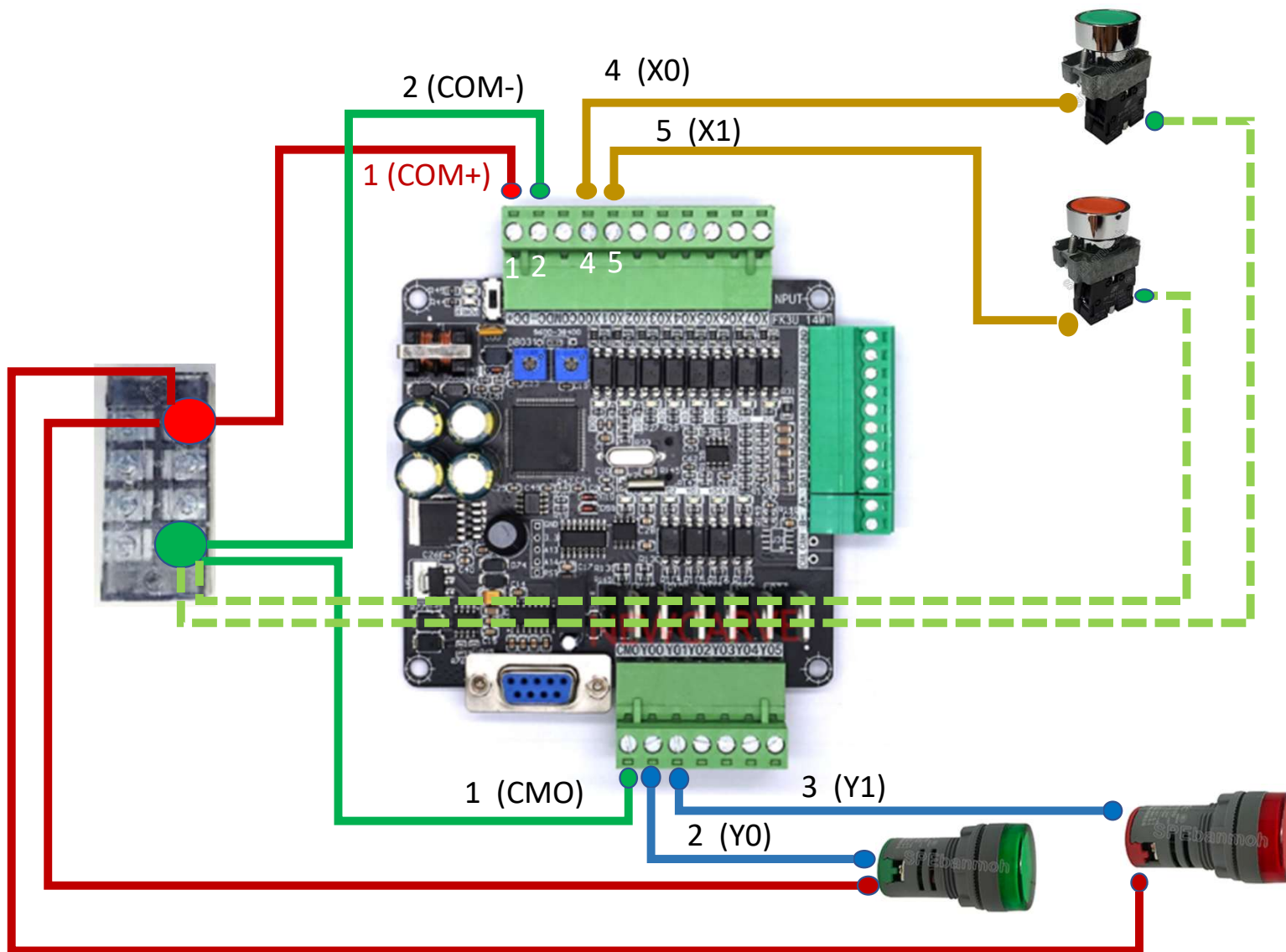
Input relay ON/OFF
ตามสัญญาณจากภายนอก

ซีควเอนซ์โปรแกรมทำงานด้วย
หน้าสัมผัสของ Input relay

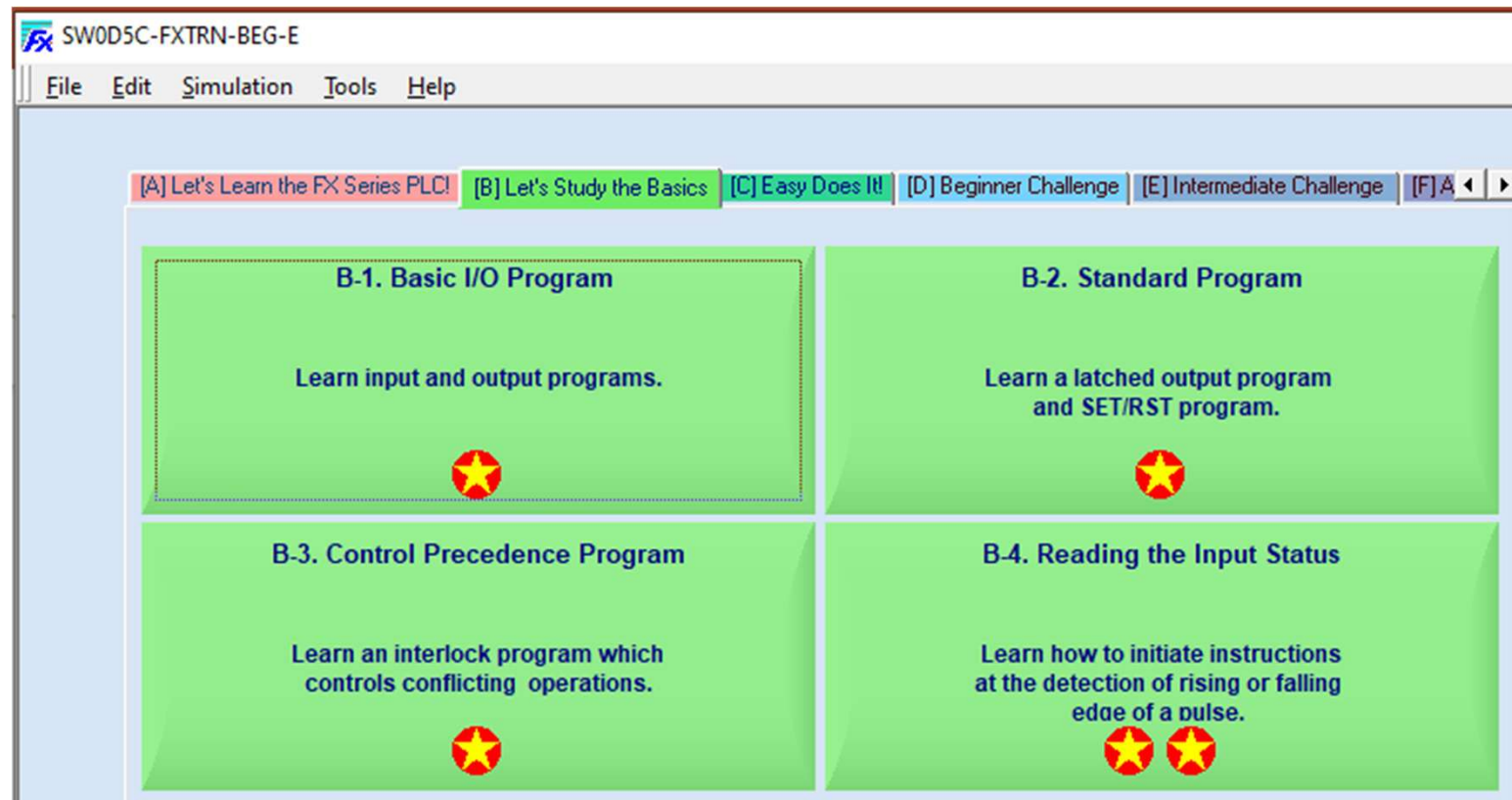
Output relay ON/OFF
ทำงานและส่งต่อ

อุปกรณ์ภายนอก Load
ทำงานหรือเคลื่อนที่





FXTRN



PLC 01 – NO NC Contact [youtube](#)

SW0D5C-FXTRN-BEG-E

File Edit Simulation Tools Help

click

Edit Ladder
Write to PLC
Reset
F T S
Main
RUN

Operating Y0
Stop Y1
Error Y2

Basic I/O Program

```
0 X020 | | (Y000 )  
2 X021 | / | (Y001 )
```

Project Edit Convert View Online Tools

X020
0 | | (Y000)
X021
2 | / | (Y001)
4 | | [END]

Write data to the PLC

RUN

Operation Panel

Lamp display

PL1 PL2 PL3 PL4
Y20 Y21 Y22 Y23

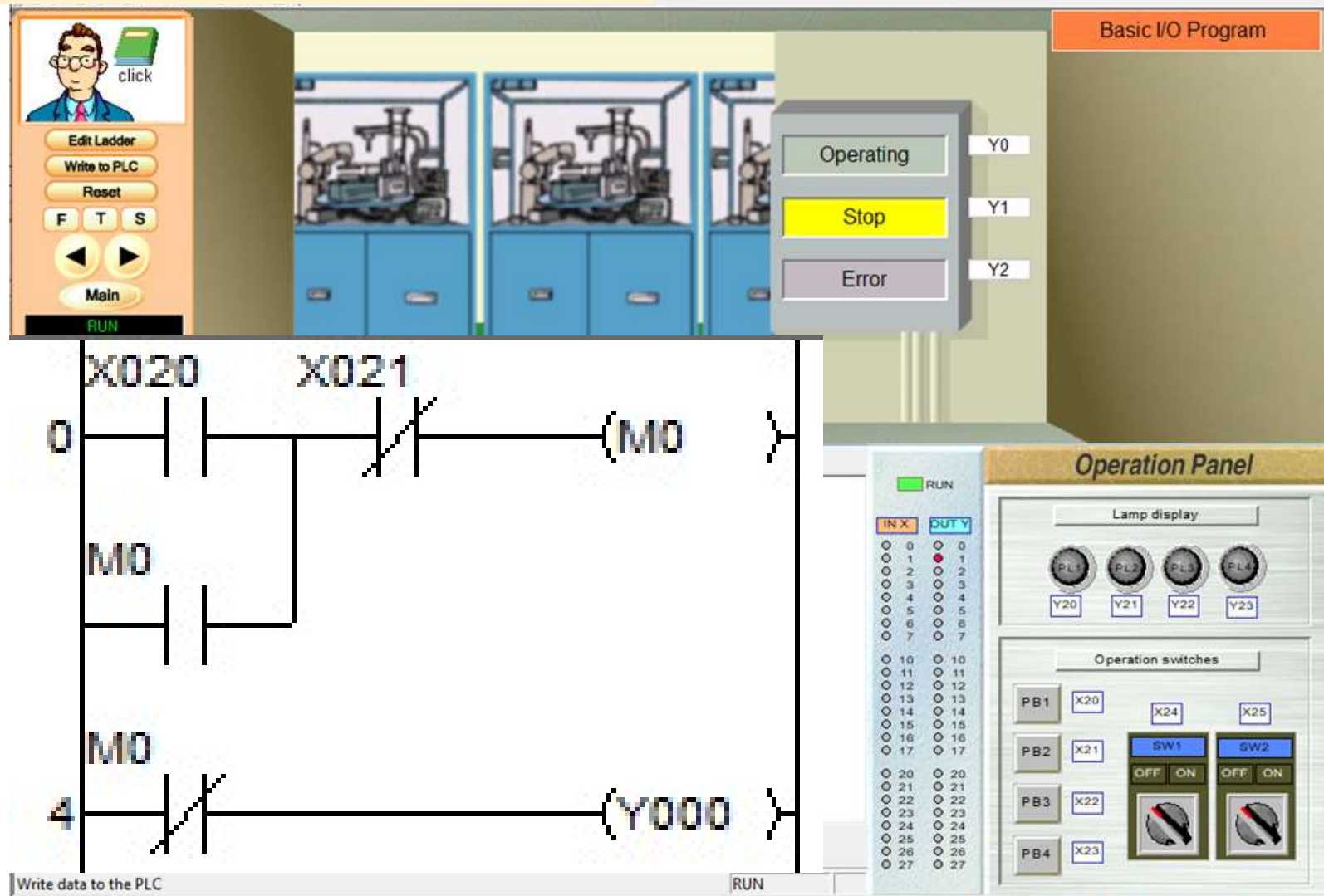
Operation switches

PB1 X20 X24 X25
PB2 X21 SW1 SW2
PB3 X22 OFF ON OFF ON
PB4 X23

IN X OUT Y

0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
10	10	10	10
11	11	11	11
12	12	12	12
13	13	13	13
14	14	14	14
15	15	15	15
16	16	16	16
17	17	17	17
20	20	20	20
21	21	21	21
22	22	22	22
23	23	23	23
24	24	24	24
25	25	25	25
26	26	26	26
27	27	27	27

PLC 02 – Self Holding [youtube](#)



[A] Let's Learn the FX Series PLC!

[B] Let's Study the Basics

[C] Easy Does It!

[D] Beginner Challenge

[E] Intermediate Challenge

[F] A



C-1. Basic Timer Operation

Learn the On-delay time function.



C-2. Application Timer Program -1

Learn the Off-delay time function
and the one shot timer.



C-3. Application Timer Program - 2

Learn a "flicker" program
executed by timers.



C-4. Basic Counter Program

Learn control methods using counters.



PLC 03 – Basic Timer youtube

File Edit Simulation Tools Help

click

Edit Ladder
Write to PLC
Reset
F T S
Main
RUN

3 Press the [F4] key to convert the program you have input.

Basic Timer Operation

X1(Upper limit)
Y5(Red)
Y6(Green)
Y7(Yellow)

Y0(Door up command)
PB1 X20
PB2 X21

Y1(Door down command)

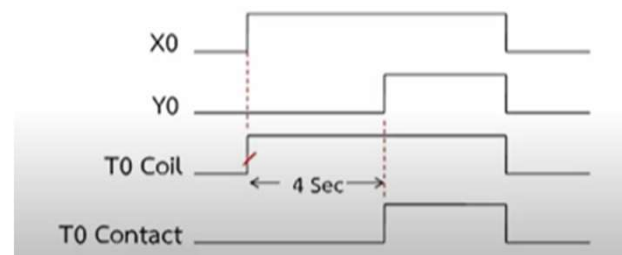
X0(Lower limit)

Timer

Enter symbol

[-()-]

T0 K30



PLC 04 – Open and Close Door [YouTube](#)

SW0D5C-FXTRN-BEG-E

File Edit Simulation Tools Help

click

Edit Ladder
Write to PLC
Reset
F T S
Main
RUN

Ladder Logic Diagram:

```
graph LR
    X020 --- Y000
    X001 --- Y000
    Y000 --- X020
    X021 --- Y001
    X000 --- Y001
    Y001 --- X021
```

Basic Timer Operation

X1(Upper limit)
Y5(Red)
Y6(Green)
Y7(Yellow)

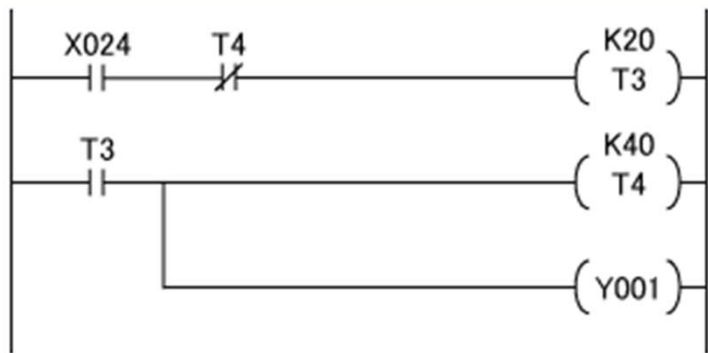
Y0(Door up command)
Y1(Door down command)

X0(Lower limit)

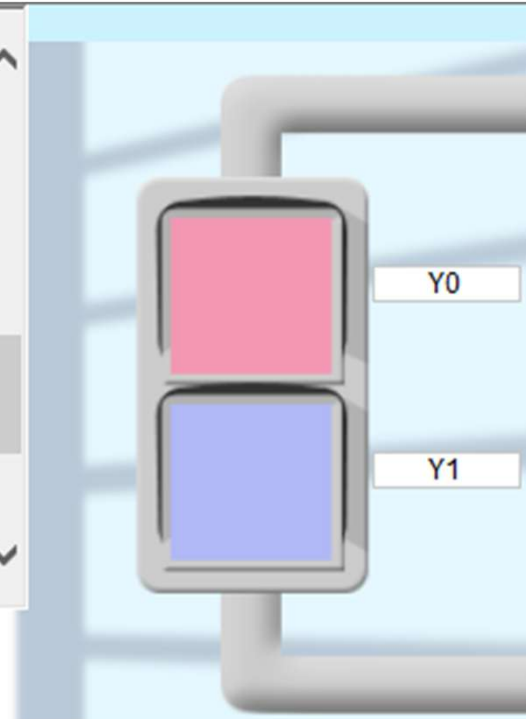
PB1 X20
PB2 X21

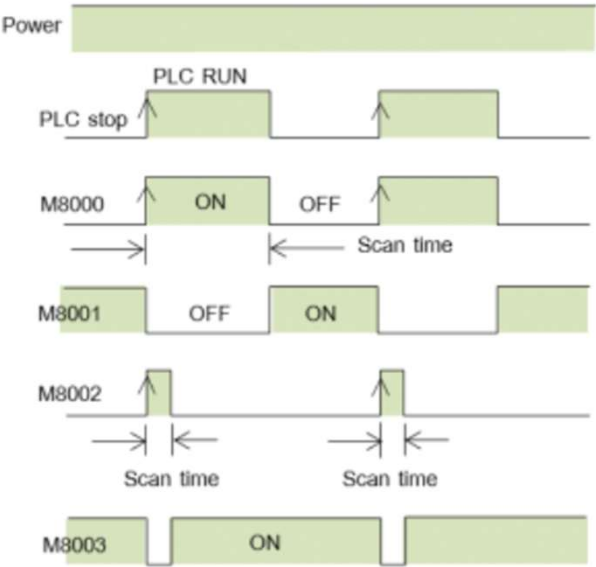
PLC 05 – Timer Cycle [YouTube](#)

In the 'Enter symbol' dialog box, enter **T3 K20** or **T4 K40** and click **[OK]** when writing the timer setting.
(Use a space between **T3** and **K20**)

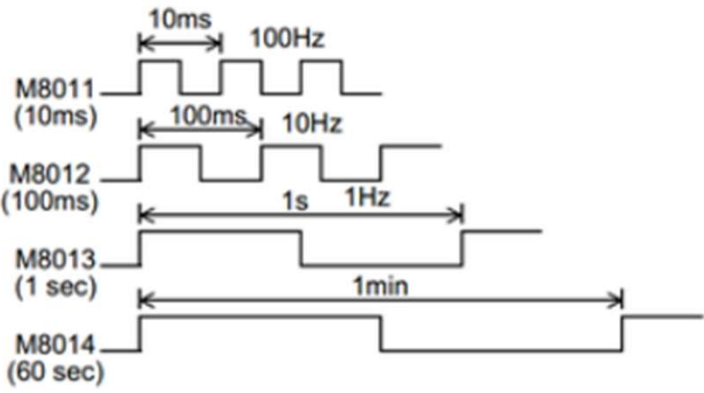


3 Press the **[F4]** key to convert the program you have input.

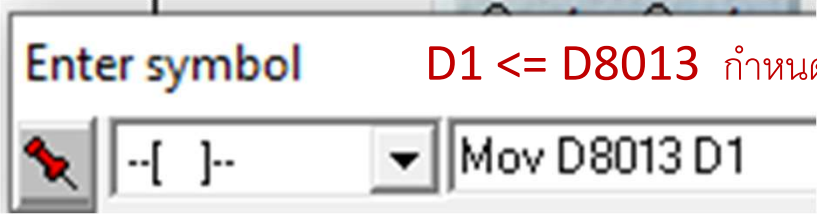




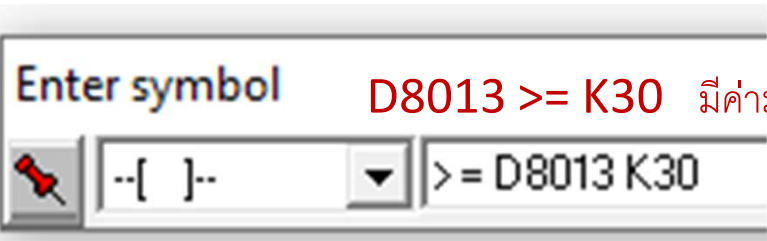
M8011 = 10 ms. (0.01 วินาที)
 M8012 = 100 ms. (0.1 วินาที)
 M8013 = 1 s. (1 วินาที)
 M8014 = 60 s. (1 นาที)



Number	Name	Set value range
D8013	Second data	0 to 59
D8014	Minute data	0 to 59
D8015	Hour data	0 to 23
D8016	Day data	1 to 31
D8017	Month data	1 to 12
D8018	Year data	00 to 99 (last two digits of year)
D8019	Day-of-the-week data	0 to 6 (which corresponds to Sunday to Saturday)



D1 <= D8013 กำหนดค่า D1



D8013 >= K30 มีค่ามากกว่า 30 หรือไม่

PLC 06 – Time Schedule [YouTube](#)

Write to PLC
Reset
F T S
Main
PROGRAM

The following timers are prepared in the special auxiliary relays for use by the PLC.
These relays can be used for simple operations such as flickering a lamp.

- M8011 10ms (0.01sec)
- M8012 100ms (0.1sec)
- M8013 1sec
- M8014 1min

ไฟสว่างระหว่าง วินาที ที่
0 – 15 และ 30 - 45

Enter symbol
--[]-- MOV D8015 D1

Enter symbol
--[]-- >= D8013 K30

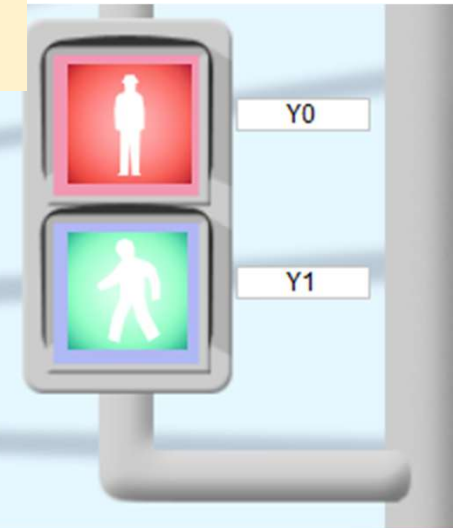
Project Edit Convert View Online Tools

M8000
2
[MOV D8015 D1]
[MOV D8014 D2]
[MOV D8013 D3]

18 [>= D3 K0] [<= D3 K15] (M1)

29 [>= D8013 K30] [<= D8013 K45] (M2)

40 M1
M2
(Y001)



Operation Panel

Lamp display

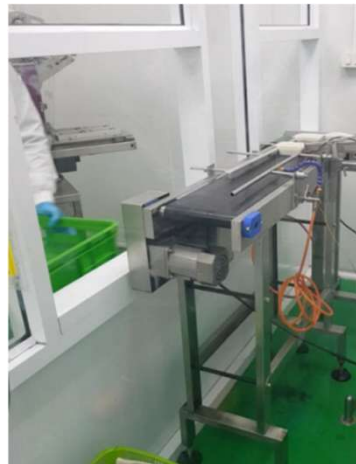
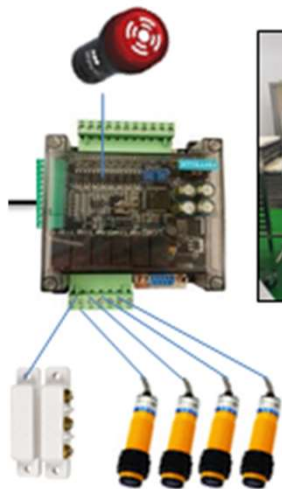
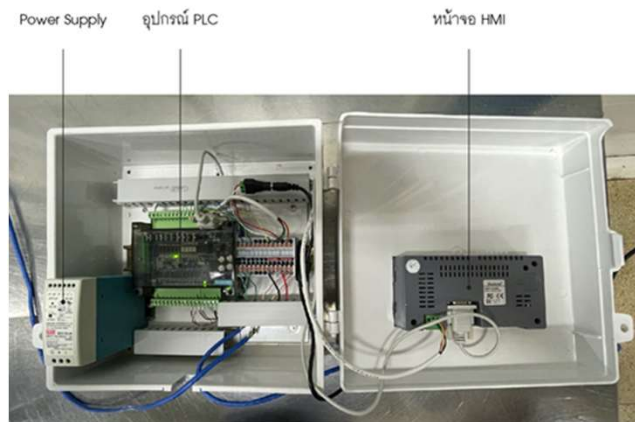
PL1 PL2 PL3 PL4
Y20 Y21 Y22 Y23

Operation switches

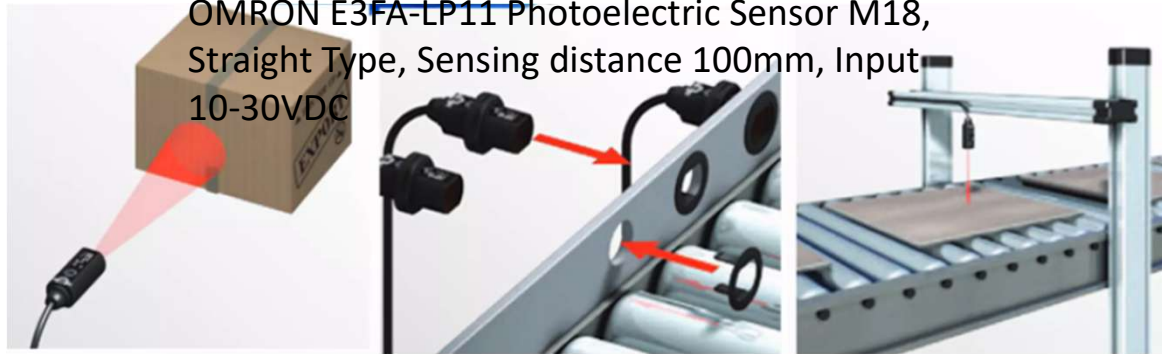
PB1 X20 X24 X25
PB2 X21 SW1 SW2
PB3 X22 OFF ON OFF ON

RUN

IN X	OUT Y
0	0
1	1
2	2
3	3
4	4
5	5
6	6
7	7
10	10
11	11
12	12
13	13
14	14
15	15
16	16
17	17
20	20
21	21
22	22
23	23



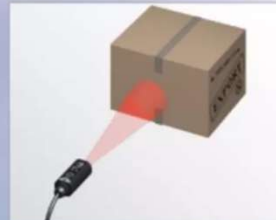
OMRON E3FA-LP11 Photoelectric Sensor M18,
Straight Type, Sensing distance 100mm, Input
10-30VDC



Unrivalled Detection with Simplicity in Setup and Installation



The short body of the E3FA/E3RA fits in tighter mounting spaces.



Visible red LED light for easy alignment.



Transparent object detection sensors utilize Omron's unique technology for detecting objects with birefringent (double refraction) properties.



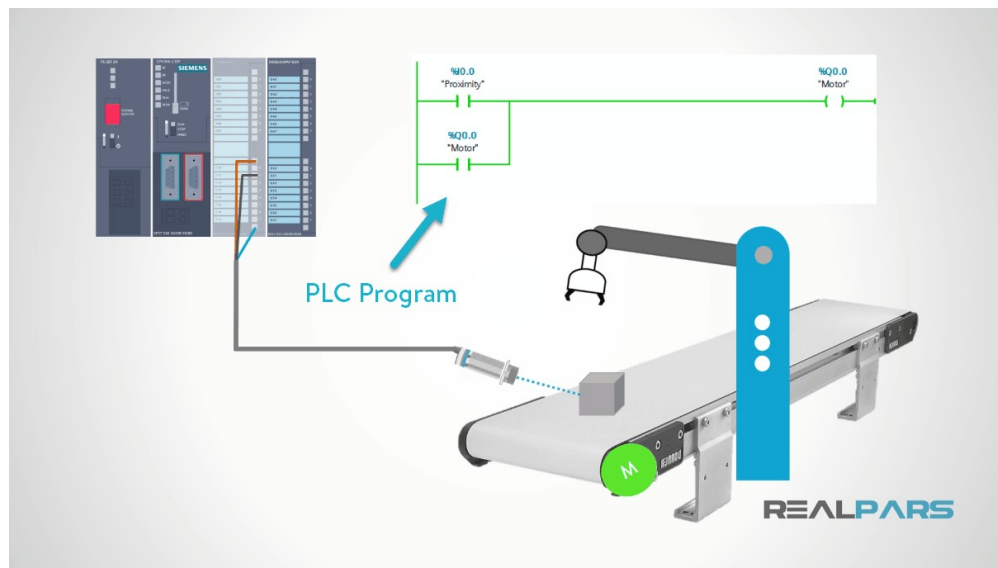
Bright LED indicators for status visibility and large sensor adjusters for use with a standard size screwdriver.



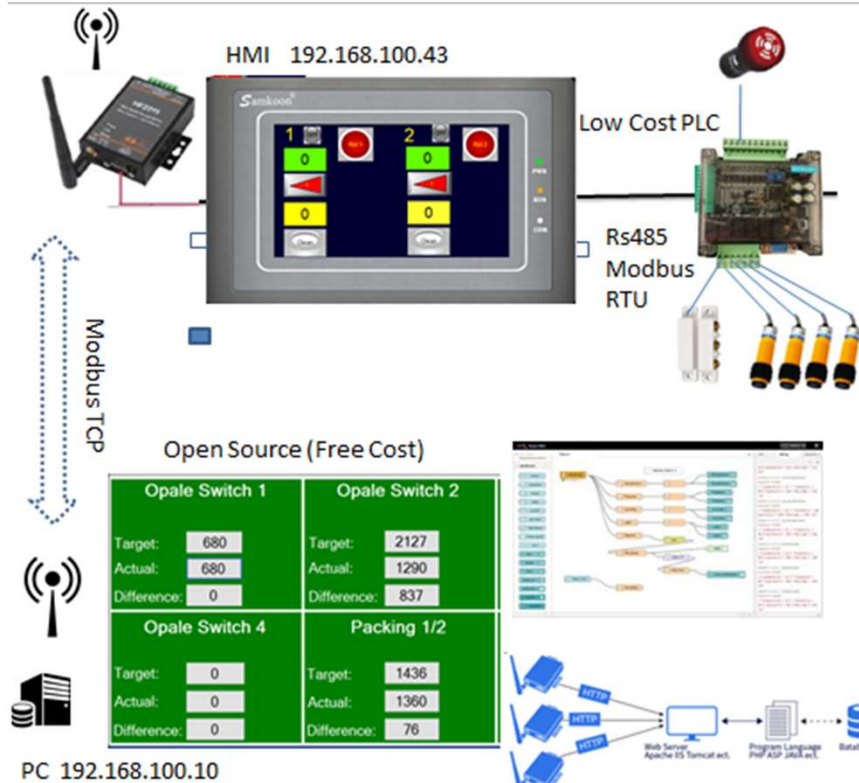
Flush, snap mounting option for quick and easy installation.



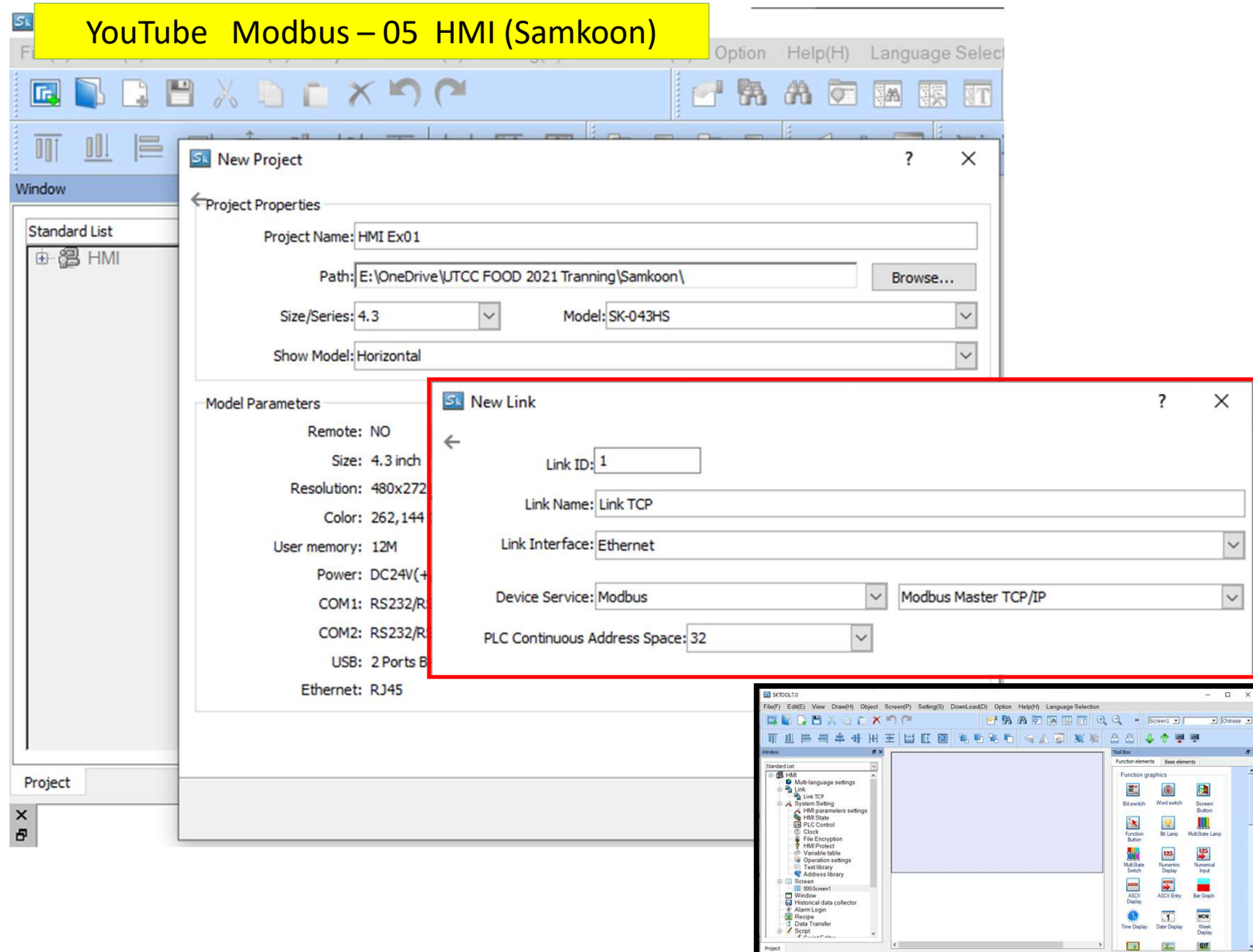
High power LED to compensate for dirt and misalignment.



ตัวอย่างการใช้ HMI (Touch Screen) ร่วมกับ PLC



YouTube Modbus – 05 HMI (Samkoon)





Bit switch

Bit Button

element type
Bit Switch

ID: BB0000

View



Prompt

Function: ON / OFF status monitoring

General Appearance Advanced Visibility

State: 1 0 Shap

Border Color:

FG Color:

BG Color:

Pattern: Solid

Function

Function: Invert

Mode: Press execute

Write Address: 0x0 Offset

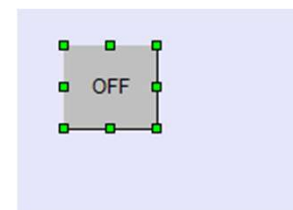
☒ Monitor ☒ Monitor Address Identical to Write A

Monitor Address: 0x0 Offset

Script

☐ Use Script

Ok



Address Entry

Standard

Link TCP

0x 0

0x
1x
3x_Bit
4x_Bit
3x
4x
3x_D
4x_D

CLR

BS

ESC

ENT

Function

Function: Invert

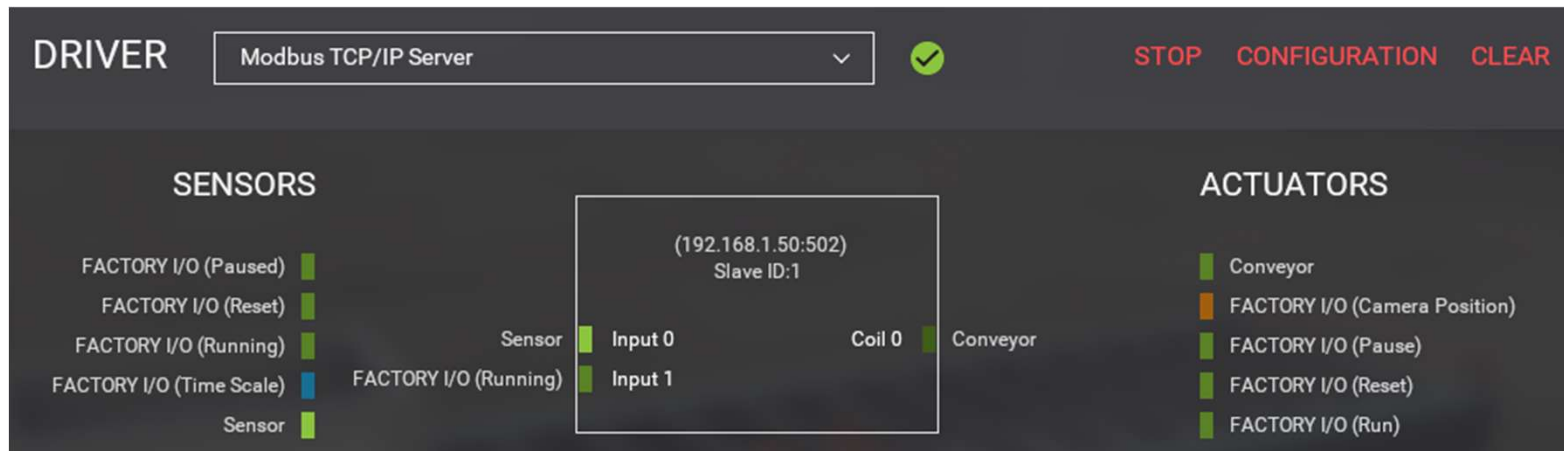
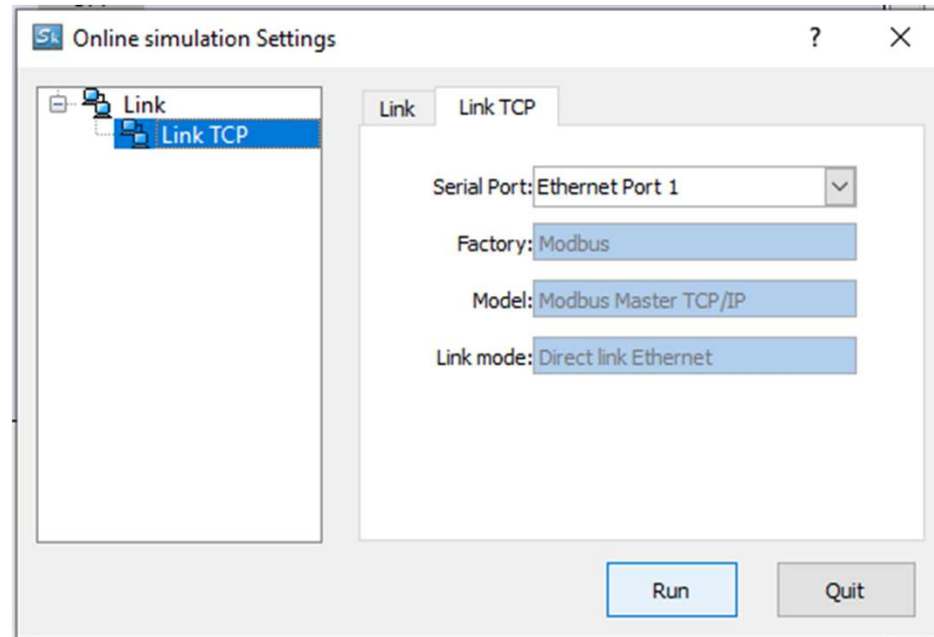
Mode: Inching
Set OFF
Set ON

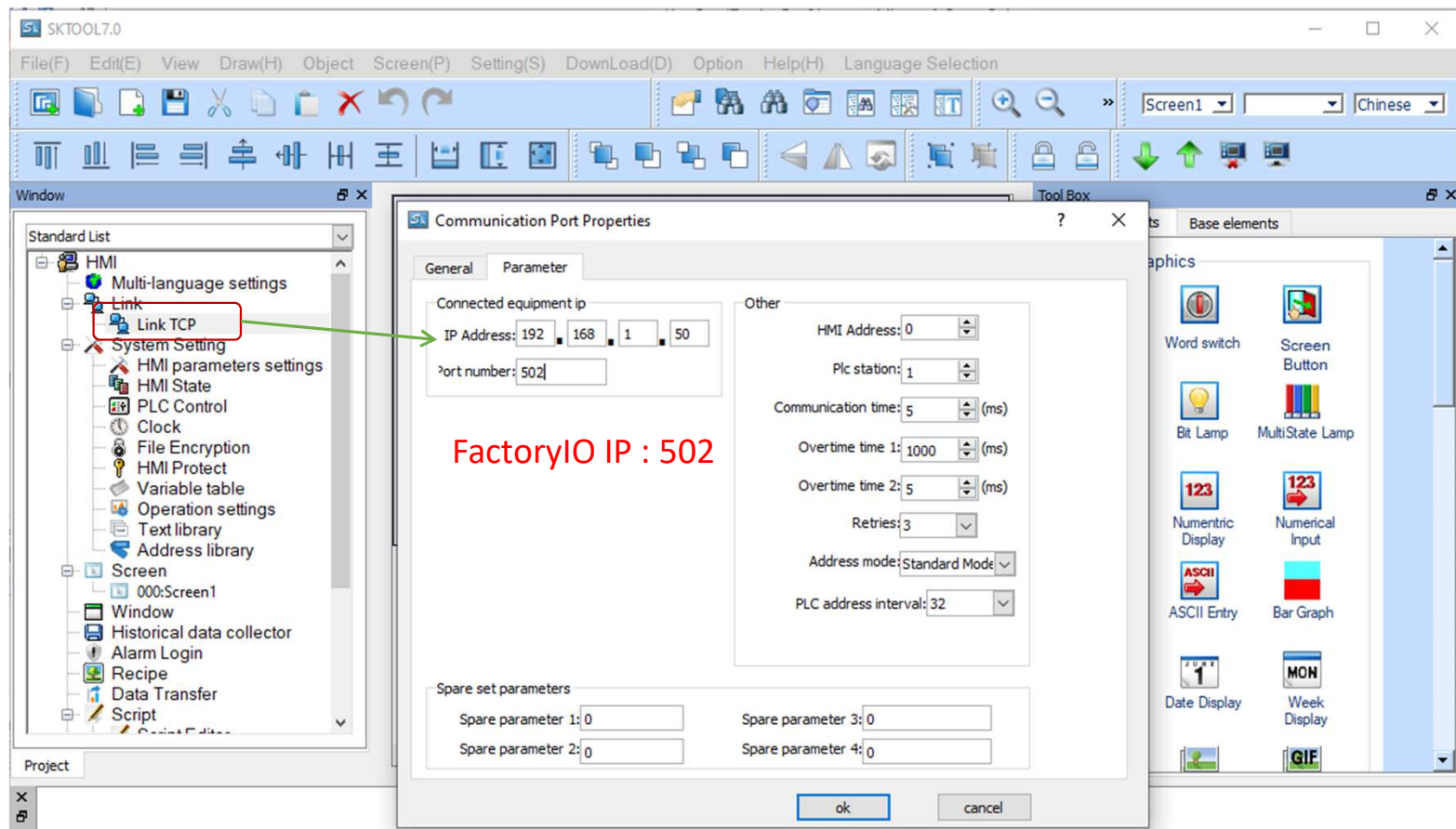
Standard

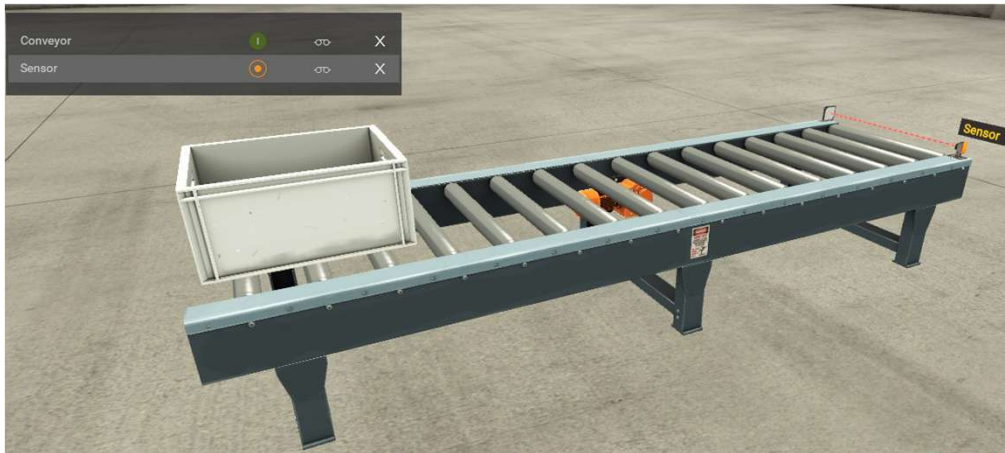
Link TCP

Internal storage area

Link TCP







DRIVER

Modbus TCP/IP Server

STARTCONFIGURATIONCLEAR

SENSORS

FACTORY I/O (Paused)

FACTORY I/O (Reset)

FACTORY I/O (Running)

FACTORY I/O (Time Scale)

Sensor

Sensor

FACTORY I/O (Running)

(192.168.1.50:502)
Slave ID:1

Input 0

Input 1

Coil 0

Conveyor

ACTUATORS

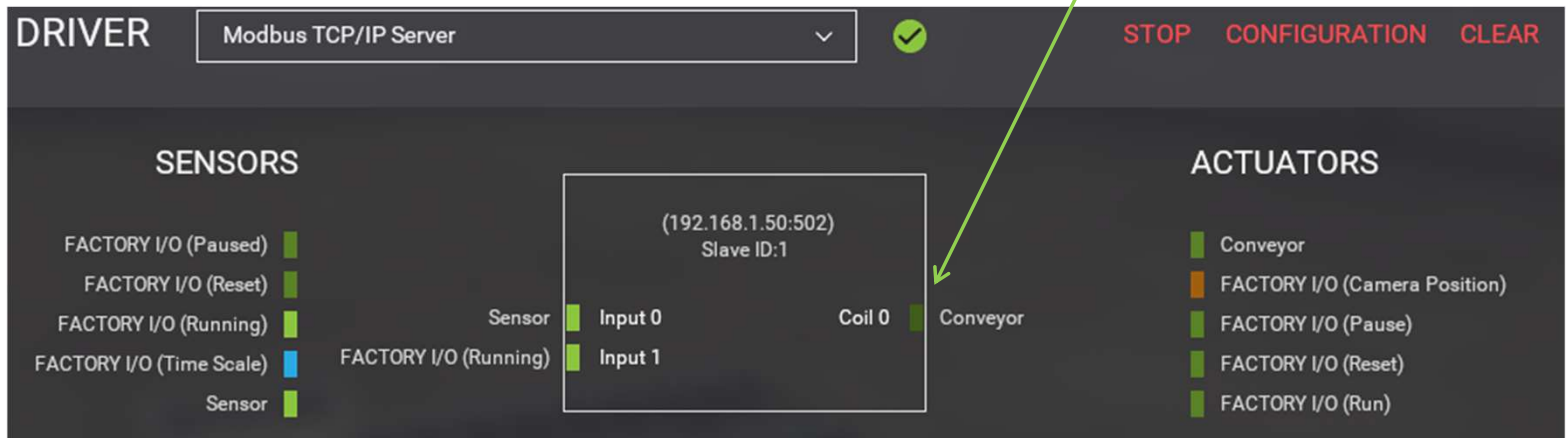
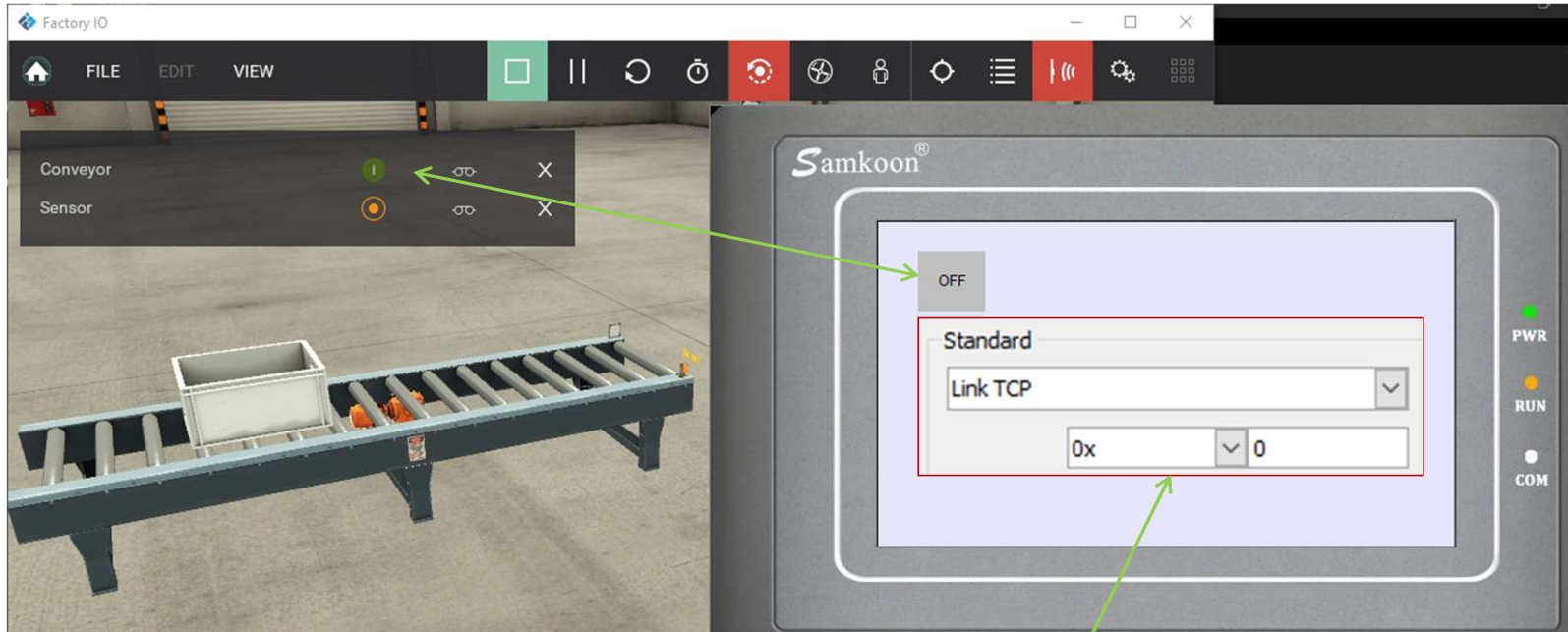
Conveyor

FACTORY I/O (Camera Position)

FACTORY I/O (Pause)

FACTORY I/O (Reset)

FACTORY I/O (Run)



Conveyor

Sensor

X

X



Samkoon®

Coil 0 Press On/Off

Input 0 Status

OFF

OFF

Standard

Link TCP

1x

0



Bit Lamp

DRIVER

Modbus TCP/IP Server

✓

STOP

CONFIGURATION

CLEAR

SENSORS

FACTORY I/O (Paused)

FACTORY I/O (Reset)

FACTORY I/O (Running)

FACTORY I/O (Time Scale)

Sensor

Sensor

FACTORY I/O (Running)

(192.168.1.50:502)

Slave ID:1

Input 0

Input 1

Coil 0

Conveyor

ACTUATORS

Conveyor

FACTORY I/O (Camera Position)

FACTORY I/O (Pause)

FACTORY I/O (Reset)

FACTORY I/O (Run)

